Detailed solutions.

Trial Examination 1

Question 1 (B) [Go to **Model Solution**]

This question tests your understanding of what the phrase **"carbon-neutral"** means.

This term refers to fuels whose combustion is associated with near-zero **net** CO_2 emissions. In other words, the amount of CO_2 released by the combustion of the carbon-based fuel is roughly equivalent to the amount of CO_2 absorbed recently from the atmosphere during the production of the fuel. **Biofuels are considered to be carbon neutral, whereas fossil fuels are not, making the answer (B).** Let's explain where this assertion comes from.

Most fuels contain carbon atoms as part of their chemical formula, for example:

- coal seam gas – primarily composed of methane (CH_4)
- coal – a mixture of extremely large and complex hydrocarbons
- crude oil – a mixture of smaller hydrocarbons such as octane (C_8H_{18})
- biodiesel – a mixture of fatty acid methyl esters

Think about how these fuels were naturally made.

Fossil fuels bear this name because they were naturally produced over millions of years through fossilisation of plant material. If we moved back in time to millions of years ago, you would have witnessed these plants growing in size and gaining mass as a result of "fixing carbon" from the atmosphere. Atmospheric carbon dioxide (CO_2) is converted to glucose ($C_6H_{12}O_6$) via **photosynthesis**, which plants subsequently convert to lots of different types of extremely large molecules (eg. complex carbohydrates and proteins). These large molecules may be "passed on" to living organisms such as animals and fungi, which rely on consumption of organisms to obtain their source of organic molecules.

On rare occasions, these organisms (plants or otherwise) may be rapidly buried (due to some natural disaster, example) in conditions that prevent decomposition (eg. burial underwater, poor oxygen concentrations in the immediate surroundings). If these dead organisms are subjected to extremely high pressures as a result of their rapid burial, multiple chemical changes occur in the large carbon-based molecules that form the plant's structure. Grossly simplifying, these carbon-based molecules lose most of their oxygen and nitrogen atoms, and some may decompose into smaller molecules. What you end up with is a set of hydrocarbons of highly variable sizes, ranging from extremely small (eg. methane - CH_4) to extremely large (over 30 carbon atoms) molecules. Mixtures containing very large hydrocarbon molecules, that exhibit really strong dispersion forces and therefore would be solid at room temperature, form solid deposits which we know as **coal**. Mixtures containing moderate-sized hydrocarbon molecules, whose dispersion forces are strong enough to render the mixture as a liquid at room temperature, form a liquid which we know as **crude oil**. Small hydrocarbon molecules such as methane may form gaseous "bubbles" under the ground on their own (natural gas) or above a crude oil deposit (natural gas again - although we can call this variant "associated gas"), or they may be adsorbed ("stuck") onto the large hydrocarbon molecules forming coal via dispersion forces. This adsorbed gas can be extracted by drilling a hole into a deposit of coal and aspirating out; we call this adsorbed gas **coal seam gas**.

Today, when we extract these fossil fuels and combust them, releasing carbon dioxide (CO_2), we are effectively "closing the loop" and returning the CO_2 back to the atmosphere, the CO_2 that plants have absorbed for photosynthesis millions of years ago. This is part of what you might know as the **carbon cycle**.

Given that the Industrial Revolution occurred only about 200 years ago, the original absorption of CO_2 millions of years ago is effectively irrelevant, however. For all intents and purposes, we are basically digging up carbon-based compounds, combusting them, and adding CO_2 to the atmosphere. We never actually "produced" these fuels manually.

How are biofuels any different? The difference is that **we have to grow crops (that photosynthesise) to generate the biofuels**. For example, we grow sugar cane to produce bioethanol, and we grow vegetables to produce biodiesel.

In other words, we produce carbon-based fuels which we combust, releasing CO_2 into the atmosphere, but the photosynthetic events that caused the absorption of CO_2 by the plants/crops from the atmosphere did not happen millions of years ago; these events happened recently, only a matter of years, when we planted the crops in the first place!

When a biofuel is combusted, and CO_2 released, we know that this CO_2 must have been absorbed from the atmosphere a matter of years ago, when the crop that gave rise to the biofuel was planted and growing. Hence, the **net** effect is that there is a near-zero release of CO_2 into the atmosphere. This is why biofuels are considered carbon neutral.

Question 2 (B) [Go to **Model Solution**]

Let us calculate the oxidation number of chromium in each of the options:

A. CrO_4^{2-}

> **Remember!**
> Where two atoms are bonded to one another covalently, the more electronegative atom takes the negative oxidation number.

> **Remember!**
> Oxygen is an extremely electronegative atom - in fact, only fluorine (F) has a higher electronegativity.

Since oxygen can form two bonds, it will have an oxidation number of -2.

Also remember that the sum of the oxidation numbers of all the atoms should equal the overall charge of the species, which on this occasion is also -2.

Hence, if x was the oxidation number of chromium, we can say that

$$x + (-2) \times 4 = -2 \Rightarrow x - 8 = -2 \Rightarrow x = 6$$

The oxidation number of chromium is $+\mathbf{6}$.

B. Cr_2O_3

By similar conventions, the oxidation number of oxygen here is -2, and the overall charge of the species has to be 0.

Hence, if x was the oxidation number of chromium, we can say that

$$2x + (-2) \times 3 = 0 \Rightarrow 2x - 6 = 0 \Rightarrow x = 3$$

The oxidation number of chromium here is $+\mathbf{3}$.

C. $H_2Cr_2O_7$

> **Remember!**
> For a non-metal, hydrogen is an extremely electropositive atom (atom with an unusually low electronegativity).

The hydrogen here will be bonded to an oxygen atom, not the metallic chromium atom (this is basically the species you get when you donate two protons to the dichromate $Cr_2O_7{}^{2-}$ ions). Since hydrogen can only form one covalent bond, it will adopt an oxidation number of +1.

By similar conventions to the previous two cases, the oxidation number of oxygen here is -2, and the overall charge of the species has to be 0.

Hence, if x was the oxidation number of chromium, we can say that

$$2 + 2x + (-2) \times 7 = 0 \Rightarrow 2x - 12 = 0 \Rightarrow x = 6$$

The oxidation number of chromium is $+\mathbf{6}$.

D. CrO_3

By similar conventions to the previous three cases, the oxidation number of oxygen here is -2, and the overall charge of the species has to be 0

Hence, if x was the oxidation number of chromium, we can say that

$$x + (-2) \times 3 = 0 \Rightarrow x - 6 = 0 \Rightarrow x = 6$$

The oxidation number of chromium is $+\mathbf{6}$.

Therefore, the answer is B.

Question 3 (B) [Go to **Model Solution**]

Remember!
In a fuel cell, remember that **oxygen gas** always **oxidises**[a] the **fuel**.
Therefore, in a fuel cell, oxygen gas always becomes **reduced**.

[a] It should be clear where the word "oxidise" came from here.

Remember!
Remember the mnemonic:

AN OIL RIG CAT

"**an**ode **o**xidation **i**s **l**oss, **r**eduction **i**s **g**ain **cat**hode"

This is a way of remembering that:

- The **definition** of an **anode** is "an electrode where **oxidation** occurs, where a chemical is being **oxidised**".
- The **definition** of a **cathode** is "an electrode where **reduction** occurs, where a chemical is being **reduced**".

Since oxygen is being reduced, it must react at the **cathode**.
The next question is - is this the positive or negative electrode? This sort of nomenclature is a way of expressing the "natural tendency of electrons to deviate from one half cell to the other half cell". This is a galvanic cell setup, where the electrons, on their own, move from one half cell to the other half cell. We say, by convention, that the electrode that the electrons (being **negatively** charged) are naturally "repelled" from (i.e. the electrode that the electrons have a natural tendency to leave) is labelled the **negative** $(-)$ electrode, consistent with the notion that like charges repel. In a fuel cell, which is a type of galvanic cell, electrons leave the anode **spontaneously** (without being forced by an external current), and therefore the anode is labelled as the negative electrode. In a similar vein, the electrode that the electrons are naturally "attracted" to (i.e. the electrode that the electrons have a natural tendency to travel towards) is labelled the **positive** $(+)$ electrode. In a fuel cell, electrons enter the cathode **spontaneously** and therefore the cathode is given a **positive** polarity.
In short, in a **galvanic cell (including fuel cells)**,

- the anode is the negative electrode
- the cathode is the positive electrode.

Hence, oxygen is being reduced at the cathode, which is the **positive** electrode. Hence, B is the answer.

Question 4 (D) [Go to **Model Solution**]

This is a fuel cell, which is a type of galvanic cell. Here, we are using a **fuel**, which is defined as a chemical that can be reacted with another chemical, leading to the **release** of energy. Therefore, we are using the chemical energy in the fuel and releasing it in the reaction, converting it in this case to **electrical energy**. Therefore, D is the answer.

Question 5 (D) [Go to **Model Solution**]

Remember that a **reductant** is a chemical whose function is to **reduce OTHER chemicals**. In other words, if it is reducing other chemicals, it is forcing other chemicals to gain electrons - which means it needs to be shoving its own electrons down the throats of these other chemicals. Since it's donating its own electrons to shove down the other chemical's throat, it itself is being oxidised. The **reductant** itself undergoes an **oxidation** reaction.

Remember!
In a reaction, a chemical is acting as the **reductant** if it is reducing the other chemical (i.e. giving electrons to the other chemical) and, in doing so, itself gets oxidised.

When you look at the electrochemical series, you can see that it is a series of **reduction** reactions. Therefore, the oxidation reactions must be the **reverse** reactions. The reductants, being the reactants in the oxidation reactions, must be the chemicals on the **right hand side of the electrochemical series.**

Remember!
The golden rule of thumb in the electrochemical series is this:

- The further **up** in the electrochemical series you go, the further in the **forward** direction the reduction reaction as written tends to go, meaning
  - the substance on the **left** (the oxidant) becomes increasingly more reactive.
  - the substance on the **right** (the reductant) becomes less and less reactive.
- The further **down** in the electrochemical series you go, the further in the **backward** direction the reduction reaction as written tends to go, meaning
  - the substance on the **right** (the reductant) becomes increasingly more reactive
  - the substnace on the **left** (the oxidant) becomes less and less reactive

Hence, the strongest **reductant** would be found on the **bottom** on the electrochemical series on the **right**.

Of the options listed, D is the answer.

Question 6 (D) [Go to **Model Solution**]

To determine what initially happens to the rate of the forward reaction, you need to consider the factors that affect the rate of reaction:

- temperature
- concentrations of reactants
- activation energy

The temperature is kept constant throughout, and the activation energy is a property of the reaction itself, so it remains the same. However, what happens when the volume of the vessel decreases?

The concentration of $[H_2]$ and $[I_2]$ increases initially and instantaneously, hence the rate of the forward reaction increases initially and instantaneously. This rules out options **B** and **C**.

What happens subsequently? To answer this question, we need to see what happens to $[H_2]$ and $[I_2]$ after their initial instantaneous increase. From here, the only thing that could change their concentrations is if the position of equilibrium changes. To see what happens to the position of equilibrium, we will employ Le Chatelier's Principle.

> **Remember!**
> Le Chatelier's Principle states that an equilibrium system will partially oppose any change imposed upon it.

We decreased the volume of the system, hence we increased the total concentration of gas molecules (and also increased the pressure) initially. The system would attempt to partially oppose this by favouring the reaction that produces fewer gaseous particles (and decrease the pressure in the vessel). The issue is that there are 2 gaseous particles on the LHS and 2 on the RHS of the equation, hence the system cannot oppose this. Thus, the equilibrium position does not change.

This means that $[H_2]$ and $[I_2]$ subsequently do not change. Therefore, the forward reaction rate subsequently remains constant at that high value after the initial instantaneous increase. The answer is thus option **D**.

Question 7 (D) [Go to **Model Solution**]

In the explanation to the previous question it was explained that $[H_2]$ initially increases instantaneously, then remains constant at that higher value. Hence, the answer is **D**.

Question 8 (B) [Go to **Model Solution**]

This question is primarily recall. You are expected to know that a catalyst provides an alternate reaction pathway with lower activation energy, meaning option (B) is correct.
Going through the other options:

A. It shifts the position of equilibrium forward in a reversible reaction.

Catalysts do **NOT** change the equilibrium position or the value of K (equlibrium constant). A catalyst increases the rate of both the forward and back reactions equally.

C. It increases the rate of reaction by lowering the activation energy of the reaction.

This option is the next best option, but is not as specific as option (B).

D. It increases the spontaneity of a reaction.

The **spontaneity** of a reaction refers to, in essence, the value of the equilibrium constant. The greater the value of K, the greater the spontaneity of the reaction.

As already mentioned, adding a catalyst does not change the value of K and, therefore, does not affect the spontaneity of the reaction.

Question 9 (D) [Go to **Model Solution**]

Remember, a change in activation energy or enthalpy will not affect the equilibrium position. Activation energy affects the **reaction rate**. If the activation energy increases, the proportion of collisions between reactants above the activation energy decreases, thereby decreasing the reaction rate.
Increasing Y means the activation energy of both the forward and the back reactions increases. This means that the rate of both the forward and back reactions decrease. Hence, **D** is the correct option.

Question 10 (D) [Go to **Model Solution**]

Let us look at each option in turn:

A. The cations in the electrolyte or salt bridge flow towards the positive electrode in both galvanic and electrolytic cells.

The cations in the salt bridge and electrolyte try to prevent a build-up of charge in either half-cell. Electrons flow from the anode to the cathode via the circuit. Hence, to prevent a build-up of negative charge in the half-cell that contains the cathode, cations from the salt bridge or electrolyte will flow towards the cathode. In the galvanic cell, the cathode is the positive electrode but in the electrolytic cell, the cathode is the negative electrode. Hence, option A is not true.

B. The primary energy transformation is from electrical to chemical in both galvanic and electrolytic cells.

The primary energy transformation is from electrical to chemical in an electrolytic cell – electrical energy is used to drive a non-spontaneous reaction. However, in a galvanic cell, energy released from a chemical reaction to drive a current. Hence, the energy transformation in a galvanic cell is from chemical to electrical, not electrical to chemical. Hence, option B is incorrect.

C. The reactants are continuously supplied in all galvanic and electrolytic cells.

The reactants are only continuously supplied in fuel cells, a type of galvanic cell. Not even all galvanic cells have a continuous supply of reactants. Hence option C is incorrect.

D. In a secondary cell, the positive electrode is always the same electrode irrespective of whether the cell is being discharged or recharged.

Let us take one electrode in a secondary cell, suppose it is the positive electrode during discharge. This electrode will be the cathode. Now, when we recharge the cell via electrolysis, this electrode, which was the site of reduction during discharge, will be the site of oxidation as the reverse reaction is occurring. This electrode will therefore be the anode. In an electrolytic cell, the anode is the positive electrode. Hence, during recharge, the electrode is still positive. Hence, option **D** is correct.

Question 11 (C) [Go to **Model Solution**]

We need to determine:

- the mass of zinc deposited on the electrode

We know:

- current passed through the solution
- potential difference across the solution
- the time that the current is run through the solution

How do we use what we know to get to what we need to determine?

We are given data about the amount of charge that passes through the solution, and we have to determine the amount of substance deposited on an electrode. What is the link? The link is that we know that 1 mole of electrons carries a charge of 96500 C. Hence, if we can find the charge that is passed through the solution, we can determine the number of electrons passed and therefore the amount of Zn deposited on the electrode!

Determining the amount of charge passed through the cell, we know that the current is 0.839 A. In other words, 0.839 coulombs of charge pass through the cell in 1 second. However, the current was run for 10 minutes.

Thus, the charge passed through the cell is equal to:

$$Q = It = 0.839 \times (10.0 \times 60) = 503 \text{ C}$$

Knowing that 1 mole of electrons has a charge of 96500 C, we can determine the amount of electrons that has passed through the cell:

$$n(\text{e}^-) = \frac{503}{96500} = 0.00522 \text{ mol}$$

Knowing that it takes 2 electrons to reduce one Zn^{2+} ion in the solution to Zn, we can determine that the amount of Zn deposited on the electrode is half the number of electrons that arrives at the electrode. Hence,

$$n(\text{Zn}) = \frac{n(\text{e}^-)}{2} = 0.00261 \text{ mol}$$

Now we can determine the mass of zinc:

$$m(\text{Zn}) = 0.00261 \times 65.4 = 0.171 \text{ g}$$

Hence, the answer is **C**.

Question 12 (B) [Go to **Model Solution**]

We are asked to determine whether the electrode is the cathode or anode, and whether it has a negative or positive polarity.

Now, remember, from AN OIL RIG CAT, that oxidation occurs at the anode and reduction at the cathode.

Therefore, we need to determine whether reduction or oxidation occurred at the electrode. We know that zinc was deposited on the cathode - zinc (Zn) was produced from zinc ions (Zn^{2+}). Zn^{2+} has accepted two electrons to produce Zn, hence this is a reduction reaction. Thus, the electrode is the cathode.

As for the polarity, remember that in an electrolytic cell that the **cathode has a negative polarity**.

Hence, the answer is **B**.

Question 13 (A) [Go to **Model Solution**]

To determine which products are produced at each electrode, we need to look at the electrochemical series. Remember, the strongest oxidant will be reduced at the cathode (remember AN OIL RIG CAT). The strongest reductant will be oxidised at the anode.

The options given are based on positive and negative electrodes. The question is - which electrode is the positive electrode? This is electrolysis, so the anode has a (+) polarity. The cathode has a (-) polarity. Hence, the positive electrode is the anode and the negative electrode the cathode.

To do this kind of question, we need to identify the species in solution. These are:

$$Mn^{2+}\text{ (aq)}, I^-\text{ (aq)}, H_2O\text{ (l)}$$

Looking for half-equations involving each of these species in the electrochemical series:

$$\begin{aligned}
H_2O_2\text{ (aq)} + 2H^+\text{ (aq)} + 2e^- &\rightleftharpoons \mathbf{2\ H_2O\ (l)} && E^\circ = +1.77\text{ V} \\
O_2\text{ (g)} + 4H^+\text{ (aq)} + 4e^- &\rightleftharpoons \mathbf{2\ H_2O\ (l)} && E^\circ = +1.23\text{ V} \\
I_2\text{ (s)} + 2e^- &\rightleftharpoons \mathbf{2I^-\ (aq)} && E^\circ = +0.54\text{ V} \\
\mathbf{2\ H_2O\ (l)} + 2e^- &\rightleftharpoons H_2\text{ (g)} + 2OH^-\text{ (aq)} && E^\circ = -0.83\text{ V} \\
\mathbf{Mn^{2+}\ (aq)} + 2e^- &\rightleftharpoons Mn\text{ (s)} && E^\circ = -1.03\text{ V}
\end{aligned}$$

The strongest oxidant is found on the LHS of the reduction half-equation with the highest E° value. The strongest reductant is found on the RHS of the reduction half-equation with the lowest E° value. Hence, the strongest oxidant is H_2O, and it will be reduced to H_2 at the cathode, which is the negative electrode.

The strongest reductant is I^-, and it will be oxidised to I_2 at the anode, which is the positive electrode.

Hence, the answer is **A**.

Question 14 (C) [Go to **Model Solution**]

Chlorine can add across a C=C double bond without a catalyst. So, mixing the green-coloured chlorine and ethene (which contains a double bond) together in container B will make the chlorine add across the double bond, and the product would be colourless, so the contents of container B will decolourise.

Chlorine can perform a substitution reaction with ethane only in the presence of UV light or heat. The question gave that the containers were exposed to sunlight, so chlorine will react with ethane in container A to form HCl and a chloroalkane. Both these products are colourless, therefore the contents of container A will decolourise. Hence, option **C** is correct.

Question 15 (B) [Go to **Model Solution**]

When a substance boils (i.e. converts from a liquid to a gas), the molecules separate completely from one another in the formation of the gas. In order to do this, they need to break the bonding that holds them together, which are **intermolecular bonds**.

The stronger the intermolecular bonding, the higher the amount of vibration and movement the molecules need to break free of this bonding in the formation of the gaseous state, and therefore the higher the boiling point.

Therefore, the question is asking - which of the molecules has the **weakest** intermolecular bonding?

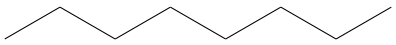
Let us draw the structures of each of the molecules:

A. octane

Breaking the name of the molecule up:

- The 'oct' affix means there are **8** carbons in the main chain. Similarly, an **oct**agon has 8 sides.
- The 'ane' affix means there are no C=C double bonds in the molecule.

Hence, the structure of the molecule is below:

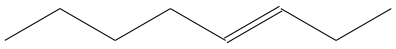


B. oct-5-ene

Breaking the name of the molecule up:

- The 'oct' affix means there are **8** carbons in the main chain. Similarly, an **oct**agon has 8 sides.
- The '5-ene' affix means there is a C=C bond on carbon-5. (Incidentally, it should be carbon-3, actually).

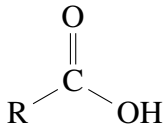
Hence, the structure of the molecule is below:



C. octanoic acid

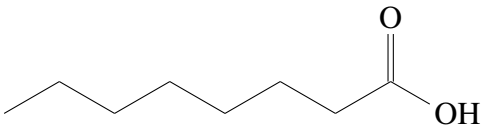
Again, the "oct" and "an" affixes mean we are dealing with an 8-carbon compound without C=C double bonds.

The affix "-oic acid", remember, refers to the carboxyl functional group:



This functional group is, always, at the end of the molecule.

The structure of the molecule is below:

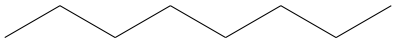


We can immediately eliminate this option given the presence of the carboxyl group, which can exhibit hydrogen bonding with one another in addition to strong dispersion forces. This option is likely to have the highest boiling point.

D. nonane

This "non" affix (as in nonagon, a 9-sided polygon), refers to a molecule with 9 carbon atoms (rather than 8).

This gives us the following:



We have already eliminated option (C). Now let's look at options (A), (B) and (D).

Now, we can eliminate D (nonane) as the answer, because nonane is a bigger molecule than options (A) and (B), and therefore would definitely not have the lowest boiling point. We are left with options A and B.

Generally, alkenes have a slightly **lower** boiling point than the equivalent alkane with the same number of C atoms. The main reason for this is that alkenes can have a "kink" in the molecule[100]. This means that these molecules would not be able to pack closely together, meaning they would be on average further away from one another in liquid form. As an analogy, take 10 pencils, bend them at the middle, and try to pack them together. You'll notice that the pencils will be more spread out than if they were not bent.

In other words, option (B) would be the correct answer.

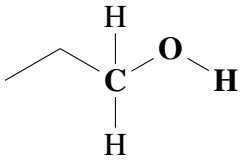
[100]This is only true of a particular isomer of this alkene called the cis- isomer. This used to be in the course but is no longer in the course. So don't stress. I included this to satisfy the pedants among us.

Question 16 (C) [Go to **Model Solution**]

Here, you can see that the Cl group is **replaced** with an OH (hydroxyl) group. This makes the reaction a **substitution** reaction.

Now, is this a redox reaction? Remember that a reaction can be classified as a redox reaction if there is a change in oxidation number to any of the atoms.
Which atoms could potentially change in oxidation number? The atoms that change the groups/atoms to which they are bonded, namely the Cl atom and the O atom.

- The O atom is originally bonded to a H as an OH^- group. The oxidation number of O here is -2 since it has an extra electron (the ion is negatively charged) and is bonded to the more electropositive H atom.
  - The O atom forms an extra bond to the C atom to form a C−O−H moiety in the product:



  - Here, the O atom is bonded to two atoms that are more electropositive than itself[101]. Therefore, its oxidation number stays as -2.
- The Cl atom is originally bonded to the C atom. Its oxidation number is -1, since it is more electronegative than the C atom.
  - Remember that in this kind of substitution reaction, when the Cl is kicked out of the molecule, it becomes a Cl^- ion; the Cl atom here still has an oxidation number of -1.

Therefore, there is no change in oxidation number in any of the atoms and hence this is not a redox reaction. Thus, the answer is C.

Question 17 (B) [Go to **Model Solution**]

Remember that the **atom economy** of a reaction is a measure of how 'green' a chemical process is - how much waste is produced. It is basically the percentage of the reagent material that is present in the product. In the context of VCE Chemistry, we **generally** ignore stoichiometry and assume every rectant and product is in a 1:1 mole ratio to one another. The formula for atom economy is therefore:

$$\text{atom economy} = \frac{\text{sum of molar masses of } \textbf{all useful products}}{\text{sum of molar masses of } \textbf{all reactants}}$$

Let us establish what the reactants and products are here.

[101] Remember, oxygen is the second most electronegative element in the Periodic Table. It's kind of hard to beat oxygen in terms of electronegativity. Unless you are fluorine.

Here, 1-chloropropane (C_3H_7Cl) reacts with hydroxide ions (OH^-) to form propan-1-ol (C_3H_8O) and Cl^- ions, in the presence of an $AlCl_3$ catalyst.

Thus, the reactants are:

- C_3H_7Cl (molar mass = 78.5 g mol^{-1})
- OH^- (molar mass = 17.0 g mol^{-1})

The **useful** product is:

- C_3H_8O (molar mass = 60.0 g mol^{-1})

Note here that Cl^- is not a useful product - it can be seen as a waste product in this context, especially given it does not even appear in the shortened reaction pathway in the question stem. Additionally, $AlCl_3$ is a **catalyst** and not a **reactant**, thus we do not look at it when discussing atom economy.

Thus,

$$\begin{aligned} \text{atom economy} &= \frac{60.0}{17.0+78.5} \\ &= 62.8\% \end{aligned}$$

The answer is therefore B.

Question 18 (B) [Go to **Model Solution**]

Let us determine the context here.

An organic compound has been freshly synthesised and we need to work out its structure. It follows that the most appropriate techniques to use in identifying the compound would be those that give information as to the structure of the compound.

Mass spectrometry (option A) will give information about the molecular formula of the compound, as well as certain clues about the structure of the compound (by a careful examination of the m/z values of the fragments).

UV-visible spectroscopy (option B) gives limited information about the identity of the compound. In certain cases, the wavelength of light absorbed can give an indication of the size of the molecule (the theory is much beyond the VCE course), but in general UV-visible spectroscopy has limited use for structural elucidation of organic compounds.

IR spectroscopy (option C) will give an indication as to the presence of certain functional groups. For instance, if there is a strong absorption band at 1600 cm^{-1}, that implies that somewhere in the molecule there is a C=O group. Hence, IR spectroscopy will be quite useful in determining the presence of certain functional groups in the compound, which will assist in determining its structure.

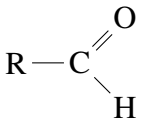
Question 19 (C) [Go to **Model Solution**]

We know the molecular formula of the compound ($C_3H_6O_2$) and it would be highly useful to work out the number of double bonds in this molecule using the double bond equivalent formula[102]:

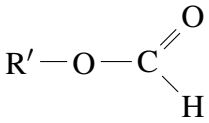
$$\text{DBE} = \frac{2[n(\text{C})+1]-n(\text{H})}{2} = \frac{8-6}{2} = 1$$

There is one double bond in this molecule (it is very unlikely to be cyclic).

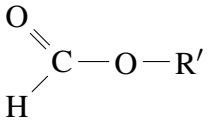
What can we work out from the NMR spectrum? We see that there is a singlet at around 8 ppm. Looking at the **Data Book**, there are a few possible groups of atoms that could give rise to a singlet around this chemical shift. However, remember we are dealing with a 3-carbon compound with no nitrogen atoms - so that removes the possibility of a benzene ring or an amide being involved here. The following group, however, seems to fit the bill somewhat (9.4-10.0 ppm):



This R group could be an R'-O group, perhaps, as below:

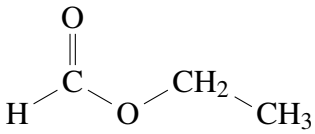


You can be astute here and pick up that this is just a **methanoate** (ester) group; to make it clearer, let's draw this molecule backwards:



That would also explain the presence of a double bond (by means of the C=O group).

There are no other double bonds in the compound, and we only have two carbon atoms to arrange, knowing we have a HCOOR structure. There is a triplet and a quartet in the NMR spectrum, meaning (by the $n+1$ rule) that one carbon is bonded to two hydrogen atoms, whereas the other is bonded to three; this means that we most likely have an ethyl (CH_3CH_2) group bonded to a carbon atom that itself has no hydrogens attached to it. Hence we get the structure ethyl methanoate:



Ester functional groups are neither acidic nor basic, and there is no acidic or basic functional group in this molecule. Hence this molecule is neither acidic nor basic, so option **C** is correct.

[102] However, the DBE formula is not necessary for VCE level - but it is nevertheless highly useful. If you do not want to use this method, you can recall that $C_3H_6O_2$ is the molecular formula for propanoic acid (C_2H_5COOH) which has 1 double bond. Hence, all isomers will have 1 double bond, as they all have the same molecular formula.

Question 20 (A) [Go to **Model Solution**]

What kind of absorption bands would this compound have in the IR spectrum? Firstly let's look at the bonds present in the molecule, because they determine the wavenumber of the absorption bands. There is the C-H bond, C-O bond and the C=O bond.

Remember (you can verify from the Data Book) that there is a characteristic stretch at 1600-1700 cm^{-1} for the C=O bond. This leads to option **A**. There would be no absorption at 1900, 2100 or 2300 cm^{-1}; the C-H stretch is at around 2900 cm^{-1} and the C-O at around 1300 cm^{-1}.

Question 21 (B) [Go to **Model Solution**]

Firstly, remember that the **retention time** (the time taken for a substance to pass through the column) identifies the compound. Hence, to be able to compare the concentration of caffeine in each of the four samples, we need to find out the retention time of caffeine so we know we are looking at the right peak in each spectrum.

To do this, we can use the caffeine standard, which we **know** contains only caffeine. As you can see, the peak here is about 102 seconds. Hence, the retention time of caffeine, under these experimental conditions, is 102 seconds. To determine how much caffeine a sample has, we need to have a look only at the peak at 102 seconds.

> **Remember!**
> Remember that the **area underneath the peak** is proportional to the **concentration of caffeine in the sample** (assuming equal volumes of all samples are added to the chromatograph).

The chromatograpm for soft drink A has a peak at 102 seconds that rises to about 6 arbitrary units. Calculating the area is impractical here, but we can sort of eyeball the area of sorts and compare to the other samples.

The chromatogram for soft drink B has a peak at 102 seconds that rises to about 40 arbitrary units and has a similar width to the corresponding peak for soft drink A. We can definitely say here that the area under this peak is much greater than that of the corresponding peak in soft drink A. Hence, we can definitely say that soft drink B has a higher caffeine content.

The chromatograph for soft drink C does **not** have a peak at 102 seconds, and therefore does not contain caffeine. There is a wide peak rising to about 45 arbitrary units, but this is at a retention time of about 110 seconds, not 102.

Soft drink B is winning so far. However, we need to compare this to a hypothetical 150 ppm caffeine standard. If we look at the original caffeine standard, which has a caffeine content of 50 ppm, we can say that the area under the corresponding peak of the 150 ppm caffeine standard would have three times the area. Even so, it is easy to eyeball that this would still be way lower than the chromatograph for soft drink B.

Therefore, B is the answer.

Question 22 (B) [Go to **Model Solution**]

The attraction between these two species are due to full positive and negative charges attracting each other, much like the way the ions in a salt crystal attract one another. This type of attraction is called an **ionic interaction**. Hence, option **B** is correct[103].

Question 23 (A) [Go to **Model Solution**]

Let us consider each statement in turn:

I H_2O will be a substrate for the protease enzyme.

As stated in the question stem, proteases 'hydrolyse proteins into their constituent amino acids'. This provides a clue; the term 'hydrolysis' means 'cutting with water', where 'hydro' means 'water' and 'lysis' means 'to cut'. Therefore, water will be a reactant and hence will be a substrate for the protease enzyme. Statement I is **TRUE**.

II This hydrolysis reaction is also known as denaturation of the protein.

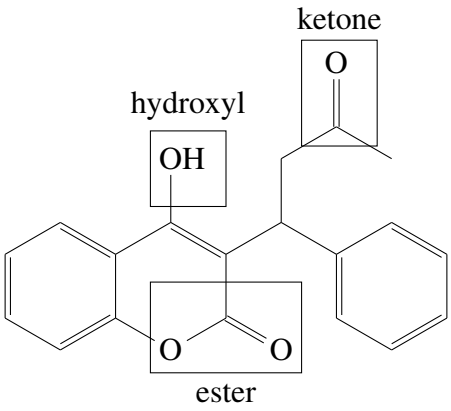
What does the term 'denature' mean? It is where the protein **changes shape** (i.e. its tertiary structure changes) as a result of an environmental temperature that is too high, or a pH that is not optimal. This is different from denaturation, where the protein is literally broken down into its constituent amino acids. Therefore, statement II is **FALSE**.

III This reaction can be considered a condensation reaction.

Remember that a condensation reaction is one where **two molecules are 'joined' together**, producing a larger molecule and a second small molecule such as water. This is not what is happening here. Therefore, statement III is **FALSE**.

Therefore, the answer is **A**.

Question 24 (D) [Go to **Model Solution**]



All three functional groups are found in warfarin.

[103]Remember that other possible interactions between amino acid residues are hydrogen bonding (for example between polar serine amino acid residues), covalent bonding (between cysteine residues) and hydrophobic interactions (for example, between alanine residues).

Question 25 (C) [Go to **Model Solution**]

Just to clarify the situation here:

- VKORC1 is the enzyme.
- Vitamin K is the substrate.
- Warfarin is the inhibitor.

Importantly, warfarin is a **reversible competitive inhibitor** of VKORC1.

> **Remember!**
> When an enzyme is **competitively inhibited** by a drug, it means that the drug can also bind to the active site of the enzyme, essentially blocking the enzyme from binding to its substrate.

> **Remember!**
> If an enzyme is competitively **and reversibly** inhibited by a drug, increasing the substrate concentration can undo the drug inhibition of the enzyme.

On this occasion, increasing the concentration of the substrate - vitamin K - will still increase the rate of production of active vitamin K (product), as the increased concentration of vitamin K will enable vitamin K to "compete" better with warfarin[104].

This makes option (C) the correct answer. Let's go through the other options:

A. ...vitamin K can react with warfarin to inactivate it.

Nothing in the question stem suggests this. Vitamin K and warfarin are alternative substrates the the enzyme VKORC1.

B. ...warfarin changes the shape of the active site of VKORC1.

Warfarin is a reversible competitive inhibitor of VKORC1. It just blocks its active site; it does not really change the shape of the active site as such.

D. ...vitamin K binds irreversibly to VKORC1.

Vitamin K is a substrate. If it bound irreversibly to VKORC1...it kind of defeats the purpose of VKORC1 converting vitamin K to active vitamin K, and the active vitamin K then being released into solution and swim around to do its job.

[104] This is actually a case of two reversible reactions competing with one another:

$$\text{vitamin K} + \text{VKORC1} \leftrightharpoons \text{vitamin K - VKORC complex} \quad (1)$$

$$\text{warfarin} + \text{VKORC1} \leftrightharpoons \text{warfarin - VKORC complex} \quad (2)$$

Increasing [vitamin K] will push equilibrium (1) forward, consuming VKORC1, which pushes equilibrium (2) backwards which is equivalent to dissociation of warfarin from VKORC.

Question 26 (D) [Go to **Model Solution**]

Reading the question leading statement:

> Complete combustion is likely to occur in...

The only other reference in the question stem about when complete combustion is likely to occur is here:

> Hydroarbons can only undergo complete combustion if **oxygen is in excess** during the reaction.

Therefore, the question stem is just asking us to figure out whether oxygen is the excess or limiting reagent in experiments I and II. If oxygen is in excess, complete combustion is likely to occur. Otherwise, it is not likely to occur.
In other words, this is an excess reagent problem. So, let's solve it.

When solving an excess reagent problem, we need to figure out:

- the amount (in mole) of **all** reactants in the reaction equation
- the mole ratio between all reactants in the reaction equation

To determine the mole ratio, we need to write down the reaction equation for the complete combustion of hexadecane in oxygen. Let's do this:

$$C_{16}H_{34} + O_2 \rightarrow CO_2 + H_2O$$

When balancing combustion equations, the usual steps I do are below:

1. Balance the C atoms first via CO_2 first.
2. Balance the H atoms via H_2O.
3. Finally, balance the O atoms via O_2.

Performing each step:

1. **Balance the C atoms first via CO_2 first.**

   The LHS has 16 C atoms as part of $C_{16}H_{34} + O_2$. So, we add a 16 next to the CO_2 to give the RHS 16 C atoms:

   $$C_{16}H_{34} + O_2 \rightarrow 16\ CO_2 + H_2O$$

2. **Balance the H atoms via H_2O.**

   The LHS has 34 H atoms as part of $C_{16}H_{34} + O_2$. Therefore, we need to put an 17 next to the H_2O (remember each H_2O has 2 H atoms) to give the RHS 34 H atoms:

   $$C_{16}H_{34} + O_2 \rightarrow 16\ CO_2 + 17\ H_2O$$

3. **Finally, balance the O atoms via O_2.**

   The RHS has 32 O atoms as part of the 16 CO_2 and 17 O atoms as part of the 17 H_2O, to a total of 49 O atoms.

   To give the LHS 49 O atoms, we need to put a $\frac{49}{2}$ next to the O_2 to give the LHS 49 O atoms[105]:

$$C_{16}H_{34} + 24.5\ O_2 \rightarrow 16\ CO_2 + 17\ H_2O$$

The mole ration of $C_{16}H_{34}$ to O_2 is 1:24.5.

Now, let's find out the actual amount of each reactant.

- **Hexadecane ($C_{16}H_{34}$)**

  The quantity of hexadecane given is 5.00 mL. We are also given its density, 0.773 g mL^{-1}.

  The given density means that 1 mL of hexadecane weighs 0.773 g.

  Since we have 5 (5.00) mL rather than 1 mL, we can quickly work out the mass of hexadecane[106]:

$$m(\text{hexadecane}) = 5.00 \times 0.773 = 3.87 \text{ g}$$

  Working out the molar mass of hexadecane[107]:

$$M(\text{hexadecane}) = 16 \times M(\text{C}) + 34 \times M(\text{H}) = 16 \times 12.0 + 34 \times 1.0 = 226.0 \text{ g mol}^{-1}$$

  and hence working out the amount, in moles, of hexadecane in this question:

$$n(\text{hexadecane}) = \frac{3.87}{226} = 0.0171 \text{ mol}$$

- **Oxygen (O_2**

  We need to do this for both experiments I and II.

  We are given **volumes** of oxygen gas here. Importantly, we are also told that the oxygen is in SLC.

> **Remember!**
> SLC means standard laboratory conditions, where the temperature[a] is 25°C and pressure is 100 kPa.
> We know that 1 mol of gas molecules, at SLC, would occupy 24.8 L of space.
> This is known as the molar volume. Just look at section 3 of the Data Book (Physical Constants and standard values).
>
> [a] I have to say, 25°C would be a little hot and stuffy in a science classroom.

  Knowing that 1 mole of O_2 gas occupies 24.8 L, and the two experiments deal with only 12.0 and 15.0 L, we can quickly "scale down" the amount of O_2 in moles in each experiment down:

$$\text{Experiment I} \qquad n(O_2) = \frac{12.0}{24.8} = 0.484 \text{ mol}$$

$$\text{Experiment II} \qquad n(O_2) = \frac{15.0}{24.8} = 0.605 \text{ mol}$$

[105] We are allowed to put fractions in reaction equations.
[106] The value in the calculator would be 3.865 g. I only rounded here to remind myself that this mass is only correct to 3 sig figs.
[107] Oftentimes, the molar mass is given to you in the question stem. I like to be mean though sometimes.

We now need to compare the amount of one of the reactants - let's pick oxygen because it is easier - against the exact amount of this reactant needed for the reaction to be complete (with neither reagent in excess).

Now, we know we have 0.0171 mol of hexadecane (see above).

We know that 1 mole of hexadecane reacts with 24.5 moles of oxygen gas. Therefore, we can find out the amount of oxygen gas needed to react with all this 0.0171 mol of hexadecane:

$$n(\mathrm{O_2})_{\text{required}} = 0.0171 \times 24.5 = 0.419 \text{ mol}$$

The amount of O_2 in both experiment I and II is greater than 0.419 mol and, hence, O_2 is in excess in both experiments. Therefore, based on the information given in the question, complete combustion would be likely to occur in both experiments.

Question 27 (B) [Go to **Model Solution**]

This question is essentially recall. Let's remember the reaction equation for photosynthesis which is essentially the reverse of cellular respiration (and the reverse of the complete combustion of glucose):

$$6\ \mathrm{CO_2}\ (g) + 6\ \mathrm{H_2O}\ (g) \rightarrow \mathrm{C_6H_{12}O_6}\ (aq) + 6\ \mathrm{O_2}\ (g)$$

Remember that photosynthesis utilises light energy[108]

Let us go through each option in turn:

A. ...involves the production of light energy by plants.

Er...nup. Light energy is used up, not produced.

B. ...utilises water as a reactant.

Yup. Just see the reaction equation. This is the answer.

C. ...leads to the production of greenhouse gases.

Oxygen is not a greenhouse gas. If anything, photosynthesis consumes greenhouse gases.

D. ...involves the oxidation of carbon dioxide.

The oxidation number of C in CO_2 is $+4$, as each O atom here would adopt an oxidation number of -2.

In glucose, if we take the oxidation number of O to be -2 and that of H to be $+1$, and the oxidation number of C to be x, we can come up with the following equation:

$$6x + 12 \times 1 + 6 \times (-2) = 6x + 12 - 12 = 6x = 0 \Rightarrow x = 0$$

Hence, the oxidation number of C in glucose is 0. The oxidation number of C has decreased from $+4$ to 0.

We can therefore say that carbon dioxide has been **reduced**, not oxidised.

[108] ...hence the term **photo**synthesis; photo- literally means "light". If you don't believe me, Google the word "photon".

Question 28 (A) [Go to **Model Solution**]

The maximum current generated in this fuel cell is 350 A. This would be the current that would maximise the rate of hydrogen fuel being consumed.

In this fuel cell, H_2 can be seen here as a reactant. In the hydrogen fuel cell, remember, hydrogen gas is reduced in either of the following reactions:

$$H_2\ (g) \rightarrow 2H^+\ (aq) + 2e^-$$
$$H_2\ (g) + 2OH^-\ (aq) \rightarrow 2H_2O\ (l) + 2e^-$$

In this case, water is being removed as a product in the O_2 side of the fuel cell (i.e. at the cathode) rather than at the anode where the H_2 is reacting. This means that the likely reaction occuring at the anode, by hydrogen, is:

$$H_2\ (g) \rightarrow 2H^+\ (aq) + 2e^-$$

The hydrogen gas deposits its electrons at the anode, and the electrons travel around the circuit to the cathode, where oxygen gas accepts the electrons. This flow of electrons is the current, the maximum of which is 350 A.

Remember!
The **current** is the rate of flow of charge, the rate at which charge flows through a circuit.
One ampere (1 A) of current means that 1 coulomb (1 C) of charge flows through a particular point in the electrical circuit every **second**.

Remember!
One mole (1 mol) of electrons carry 96,500 coulombs of charge. This 96,500 is known as Faraday's constant.

This means that 350 C of charge is being deposited on the anode every second. This charge is carried by electrons, which are deposited by the H_2. If we can find the amount of electrons being deposited on the anode in 1 hour, we can find out the amount of H_2 being consumed in 1 hour at this maximum current.

If 350 C of charge is being deposited on the anode every second, how about in 1 hour (or 60×60 seconds)?

$$Q = It = 350 \times (60 \times 60) = 1.26 \times 10^6 \text{C}$$

Knowing that 1 mole of electrons carry 96,500 C of charge, we can find out the amount of electrons deposited on the anode:

$$n(e^-) = \frac{1.26 \times 10^6}{96500} = 13.1 \text{ mol}$$

Now that we know that 13.1 mole of electrons are deposited on the anode, and that (based on the reaction half equation) 1 mole of H_2 deposits 2 moles of electrons on the anode, we can say that:

$$n(H_2) = \frac{13.06}{2} = 6.53 \text{ mol}$$

The amount of H_2 maximally consumed in 1 hour would be 6.53 mol. However, the question asks for the volume of H_2 in SLC. Knowing that 1 mole of H_2 occupies 24.8 L in SLC:

$$V = 6.53 \times 24.8 = 162 \text{ L}$$

leading to the answer (A).

Question 29 (C) [Go to **Model Solution**]

The claim here is that:

> ...hydrogen-powered buses are more **environmentally friendly** than the use of conventional diesel-powered buses...

Let us examine each option to see which of them would be a suitable argument to support this claim. Remember, the correct answer must not only be **true** in its own right; it must show that hydrogen-powered buses are more environmentally friendly.

A. ...there would be no harmful greenhouse gas emissions caused by the widescale utilisation of hydrogen-powered buses.

No harmful greenhouse gas emissions - presumably we are talking about carbon dioxide (CO_2) and other greenhouse gases that are not water[109].

In the bus itself, water vapour is produced as the product from the combustion of hydrogen, and no other gas. At initial glance, this argument appears to be correct. However, think about it more closely. Where would the hydrogen come from? It is not found naturally.

The answer to this is in the question stem:

> These buses are fitted with hydrogen tanks that could be refilled regularly at filling stations. The hydrogen gas could be sourced industrially from the steam reformation of methane derived from fossil fuel sources via the following reaction:
>
> $$CH_4\ (g) + 2\ H_2O\ (g) \rightleftharpoons CO_2\ (g) + 4\ H_2\ (g)$$

If we used hydrogen-powered buses widely, we'd have to produce hydrogen industrially to generate the fuel needed to power these buses. This hydrogen can be, as shown in the question stem, produced from reacting methane and water vapour. As the reaction equation shows, carbon dioxide (CO_2) is produced as a byproduct of the manufacture of the hydrogen fuel.

So, yes, there would be harmful greenhouse gas production from the widescale utilisation of hydrogen-powered buses. Indirectly, anyway.

So, option (A) is actually not entirely correct in its own right.

B. ...water vapour is the only exhaust gas.

This is indeed true. Water vapour is indeed the only gas produced from the combustion of hydrogen in the fuel cell. Does this make hydrogen-powered buses more environmentally friendly?

[109] Remember, water is a greenhouse gas, but the amount of water vapour the atmosphere holds at any one point is relatively constant - excess water comes down as rain. Or hail. Or snow.

From a greenhouse gas production point of view, not really. Remember that to generate the hydrogen gas itself, we have to undergo steam reformation of methane as per the reaction equation above[110], which would lead to CO_2 production. The only difference here is that in diesel-powered buses, the CO_2 is produced in the bus itself within the engine, whereas for hydrogen-powered buses where the hydrogen is sourced from the steam reformation of methane, the CO_2 is produced in the steam reformation plant rather than the bus itself.

All in all, this option is true in its own right - but would it really support the argument that hydrogen-powered buses are more environmentally?

Probably not, at least from a greenhouse gas point of view[111].

C. ...there would be fewer harmful greenhouse gas emissions caused by the widescale utilisation of these buses due to the greater energy efficiency of the hydrogen fuel cell.

This would make sense. We know, firstly, that hydrogen fuel cells are more **energy-efficient** compared to internal combustion engines - in other words, a considerably greater amount of the chemical energy released from the combustion of the fuel is converted to useful mechanical energy that is used to move the vehicle, as opposed to the less useful thermal energy.

Logically speaking, if using a hydrogen fuel cell is more energy-efficient, then you would need to use **less** of the fuel for the same amount of travel the bus undergoes - meaning less hydrogen would have to be produced from steam reforming methane, meaning that less CO_2 would need to be produced than if the hydrogen fuel-cell were of similar efficiency to an internal combustion engine[112].

This seems like a reasonably strong argument.

D. ...hydrogen is a renewable fuel.

Remember that a renewable fuel is one that can readily be replenished through natural processes.

Hydrogen gas (H_2) is synthesised from methane via the steam reforming reaction. It is not readily produced naturally. Even if we consider hydrogen gas an "intermediate compound" from an ultimate fuel - the ultimate fuel here could easily be methane if we used steam reforming to produce the hydrogen gas.

In other words, option (D) is not exactly true even in its own right.

[110] You could also electrolyse water using renewable energy to produce hydrogen gas - which would be environmentally friendly.

[111] Not in the course content, but you could argue that in diesel-powered engines, where the diesel contains some nitrogen-containing compounds and sulfur-containing compounds, combustion of diesel would lead to the production of some nitrogen oxides such as NO and NO_2, and sulfur oxides such as SO_2 and SO_3. Additionally, some of the diesel might not even be combusted but simply "crack" in the heat into smaller gaseous hydrocarbons such as ethane, or be partially oxidised to carbon monoxide (CO) or even elemental carbon (C). All of these would lead to various issues related to pollution. Using this point, you could argue that option (B) is a reasonable enough answer.

[112] Granted, we are comparing apples and oranges here - comparing hydrogen (and ultimately methane, since it is the original source of hydrogen) as a fuel to diesel as a fuel in terms of their relative CO_2 production per unit energy released. In any case, though, hydrogen fuel cells are overwhelmingly more energy-efficient (approximately 60%) than internal combustion engines (approximately 25%), which would more than overshadow any difference in CO_2 production per unit energy released between methane and diesel.

Question 30 (A) [Go to **Model Solution**]

In this procedure:

- the marshmallow is combusted, with the chemical energy in the marshmallow being released as thermal energy
- some of the thermal energy is absorbed by the water in the test tube, heating up the water
- the rest of the thermal energy is absorbed by the surrounding air - which essentially leads to "heat loss"

To work out the percentage of energy lost as heat to the surroundings, we need to work out:

- how much energy is contained within the marshmallow that can be released through combustion
- how much energy is actually absorbed by the water in the test tube

1. **Energy contained within the marshmallow that can be released through combustion**

   The question stem states that marshmallows are almost entirely carbohydrates, and that carbohydrates have an energy content of approximately 16 kJ g^{-1}. In other words, if you combust 1 g of any carbohydrate, you would release approximately 16 kJ of energy.

   In this problem, however, you only have 0.0329 g of marshmallow - i.e. 0.0329 g of carbohydrate. Therefore, the energy that can be released from the combustion of the carbohydrates in the marshmallow would be:

   $$E(\text{marshmallow}) = 0.0329 \times 16 = 0.526 \text{ kJ} = 5.3 \times 10^2 \text{ J}$$

2. **Energy absorbed by the water in the test tube**

   So, firstly:

   Remember!
   Water has a specific heat capacity of 4.18 J $°C^{-1}$ g^{-1}.
   This means that it takes 4.18 J to heat up 1 g of water by 1°C.
   So, the energy required (in J) to heat up m g of water by ΔT°C is

   $$q = m \times 4.18 \times \Delta T$$

   We use the symbol c to denote specific heat capacity, so this formula can also be expressed as:

   $$q = mc\Delta T$$

   which you would be more familiar with.

   Given we are heating 10.0 mL of water, and we have heated it from 19.5°C to 27.6°C, meaning we heated the water by

   $$\Delta T = 27.6 - 19.5 = 8.1°C$$

and knowing that the density of water[113] as per the Data Book is 1.0 g mL^{-1} meaning that 10.0 mL of water weighs 10 g, we can work out the energy absorbed by the water:

$$q = mc\Delta T = 10 \times 4.18 \times 8.1 = 3.4 \times 10^2 \text{ J}$$

Now, we can work out the percentage of energy released by the marshmallow combusting, that has been absorbed by the water[114]:

$$\text{percentage heat absorbed by water} = \frac{339}{526} \approx 64\%$$

The percentage heat loss would be 36% $(100 - 64)$, which is closest to option (A).

[113] Beware - the old Data Book for the previous Study Design in 2023 used a value of 0.997 rather than 1.0. VCAA will expect you to use the values in the Data Book for your calculations, otherwise you will lose marks.

[114] I am using unrounded values here such as 339 and 526 that you would have gotten in the previous calculations, so as to avoid rounding errors.

SECTION B - Short Answer Questions

Question 1 (a) [Go to **Model Solution**]

Energy sources are the various means by which we can obtain energy to provide electrical energy. Examples include coal, oil, solar power and wind power.

You should be very familiar with the term "renewable" which, in simple terms, means that it can be reproduced.

A renewable energy source, therefore, is an energy source that can be replenished by natural processes in a relatively short period of time (you will find that this is almost VCAA's wording in its study design although VCAA refers to renewable fuels, the term of which has a similar definition).

Question 1 (b) [Go to **Model Solution**]

Remember that the phrase "**complete** combustion" implies that the products of the combustion reaction (burning in **oxygen**) will be carbon dioxide (CO_2) and water (H_2O). The exceptions are that if carbon (graphite) is the fuel, CO_2 would be the only product since the fuel has no hydrogen atoms to form water as a product, and if hydrogen were the fuel H_2O would be the only product since the fuel has no carbon atoms to form CO_2 as a product. Let us balance the combustion equation for **methane**:

$$CH_4 + O_2 \rightarrow CO_2 + H_2O$$

In balancing combustion equations, the easiest approach is below:

1. Balance C atoms by adding the appropriate coefficient next to the CO_2.
2. Balance H atoms by adding the appropriate coefficient next to the H_2O.
3. After doing steps 1 and 2 in either order, balance O atoms by adding the appropriate coefficient next to the O_2. **You are allowed to use fractions**.
4. Add states.

Going through each step[115]:

$CH_4 + O_2 \rightarrow CO_2 + H_2O$	$\Leftarrow$ CH_4 has **1** carbon atom. There will be only 1 CO_2 molecule.
$CH_4 + O_2 \rightarrow CO_2 + 2H_2O$	$\Leftarrow$ CH_4 has **4** hydrogen atoms. There will be 2 H_2O molecules which contain a total of 4 H atoms.
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	$\Leftarrow$ There are has **4** oxygen atoms on RHS. There will be 2 O_2 molecules which contain a total of 4 oxygen atoms.
$CH_4\ (g) + O_2\ (g) \rightarrow CO_2\ (g) + 2H_2O\ (l)$	$\Leftarrow$ Final answer, with states added.

[115]Note here that water is conventionally shown as a liquid, particularly in thermochemical equations. However, past VCAA exams have accepted water as a gas (presumably due to the high temperatures generated during combustion of fuels). Importantly, water **must** be a liquid in a **thermochemical equation** unless suggested otherwise by the question; you may get away with putting water as a gas in a thermochemical equation as students have done in the past, but VCAA has every excuse/reason to start penalising students in future years.

Doing the same thing for graphite:

$$\begin{aligned} C &\rightarrow CO_2 && \Leftarrow \text{C has, well, 1 C atom. There will be only 1 } CO_2 \text{ molecule.} \\ C + O_2 &\rightarrow CO_2 && \Leftarrow \text{Carbons balanced. No hydrogens. 2 oxygen atoms on RHS, therefore } \mathbf{1} \text{ } O_2 \text{ molecule on LHS.} \\ C\,(s) + O_2\,(g) &\rightarrow CO_2\,(g) \end{aligned}$$

Question 1 (c) [Go to **Model Solution**]

What 'volume of carbon dioxide produced at SLC per MJ of energy released' means is:

If we burn enough fuel to release 1000 kJ of energy, what volume of CO_2 at SLC would be produced?

We should pick up that

- we know from the question stem that combusting **1 g** of methane and carbon releases **55.0 kJ** and **33.0 kJ** respectively
- we know a **mole ratio** between the amount of fuel combusted and the amount of carbon dioxide produced from the equations in part **b.**
- the volume of greenhouse gases is proportional to the amount (in mol) of greenhouse gases produced; 1 mole of a gas occupies 24.8 L at SLC (standard laboratory conditions).

We can use these facts to construct a method by which we can solve this question, for **each** fuel:

1. Work out the **mass** of fuel required to release **1000 kJ** of energy.
2. Work out the **amount (in mol)** of fuel from the mass of fuel calculated.
3. Using the mole ratio, work out the **amount (in mol)** of carbon dioxide (CO_2) that would be produced.
4. Work out the **volume (in L)** of CO_2 produced at SLC.

1. **Work out the mass of fuel required to release 1000 kJ of energy.**

   We know that burning **1 g** of methane will release **55.0 kJ** of energy.

   It would follow that burning **2 g** of methane would release **110 kJ** of energy (55.0×2).

   So how many grams of fuel would you need to burn to release **1000 kJ** of energy? This is the same as asking how many lots of 55.0 kJ fit into 1000 kJ:

$$\begin{aligned} m(CH_4) &= \frac{1000}{55.0} \\ &= 18.1 \text{ g} \end{aligned}$$

   So, we would need to burn **18.1 g** of methane to release **1000 kJ** of energy.

Performing a similar calculation for carbon, knowing that burning **1 g** of graphite will release **33.0 kJ** of energy:

$$\begin{aligned} m(\text{C}) &= \frac{1000}{33.0} \\ &= 30.3 \text{ g} \end{aligned}$$

So, we would need to burn **30.3 g** of graphite to release **1000 kJ** of energy.

Summarising:

$$m(\text{CH}_4)_{1000\text{ kJ}} = 18.1 \text{ g}$$
$$m(\text{C})_{1000\text{ kJ}} = 30.3 \text{ g}$$

2. **Work out the amount (in mol) of fuel from the mass of fuel calculated.**

   We can use the formula

   $$n = \frac{m}{M}$$

   to interconvert between the mass and amount of substance.

   The molar mass (M) of methane is 16.0 g mol^{-1}.

   The molar mass of carbon (graphite) is 12.0 g mol^{-1}.

$$\begin{aligned} n(\text{CH}_4) &= \frac{18.1}{16.0} \\ &= 1.14 \text{ mol} \\ n(\text{C}) &= \frac{30.3}{12.0} \\ &= 2.52 \text{ mol} \end{aligned}$$

3. **Using the mole ratio, work out the amount (in mol) of carbon dioxide (CO_2) that would be produced.**

   Looking at both equations to get our mole ratios:

$$\text{CH}_4\text{ (g)} + \mathbf{2}\ \text{O}_2\text{ (g)} \rightarrow \text{CO}_2\text{ (g)} + \mathbf{2}\ \text{H}_2\text{O (l)}$$
$$\text{C (s)} + \text{O}_2\text{ (g)} \rightarrow \text{CO}_2\text{ (g)}$$

   The mole ratio between the fuel and CO_2 for both reactions is 1:1. Hence,

$$n(\text{CO}_2)_{\text{methane}} = n(\text{CH}_4) \qquad \Rightarrow (\text{CO}_2)_{\text{methane}} = 1.14 \text{ mol}$$
$$n(\text{CO}_2)_{\text{graphite}} = n(\text{C}) \qquad \Rightarrow (\text{CO}_2)_{\text{graphite}} = 2.52 \text{ mol}$$

4. **Work out the volume (in L) of CO_2 produced at SLC.**

   We know that only **1 mole** of CO_2 occupies **24.8 L** at SLC. So, what volume would 1.14 mole or 2.52 mol of CO_2 hold?

$$V(\text{CO}_2)_{\text{methane}} = 1.14 \times 24.8 = 28.2 \text{ L}$$
$$V(\text{CO}_2)_{\text{graphite}} = 2.52 \times 24.8 = 62.6 \text{ L}$$

Now. The next part of the question asks:

> ...explain whether your answer supports the premise for using natural gas as a transitional fuel.

Let's figure this out. Let's look at the question stem.

The question stem says that:

> Natural gas is being discussed as a potential "transitional fuel" in the move away from coal towards renewable sources of energy for electricity generation...**the appeal of natural gas is that it is thought to lead to less greenhouse gas emissions per unit of energy generated from its combustion**.

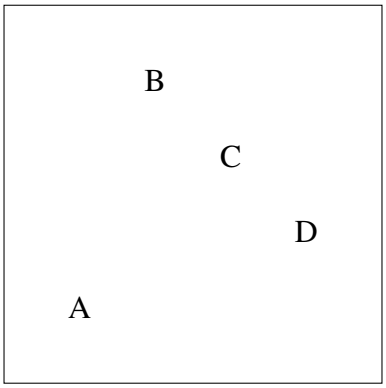
Do the numbers we calculated show this? Well, it shows that the volume of CO_2 produced per unit energy released in the combustion of methane is much lower than that of graphite, so definitely!

Question 2 (a) [Go to **Model Solution**]

Let us take a hypothetical reaction between gaseous reactants A and B, with products being C and D, like so:

$$\text{A (g)} + \text{B (g)} \rightleftharpoons \text{C (g)} + \text{D (g)} \qquad (1)$$

Let us mix these four gases in a vessel:



What the above reaction equation means is that, in this vessel, there are **two** reactions occurring:

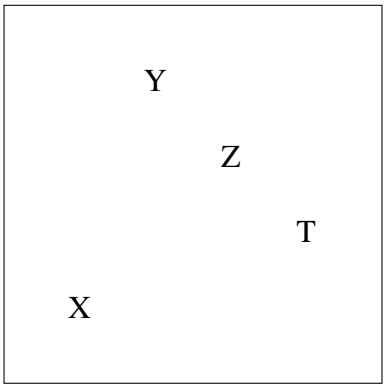
$$\text{A (g)} + \text{B (g)} \rightarrow \text{C (g)} + \text{D (g)} \qquad (2)$$
$$\text{C (g)} + \text{D (g)} \rightarrow \text{A (g)} + \text{B (g)} \qquad (3)$$

As you can see, reactions (2) and (3) are just reaction (1), just going in either direction. Here, we say reaction (1) is **reversible**. This means that this reaction can occur in the forward and back directions simultaneously at significant rates.

Let's consider an analogous reaction:

$$\mathrm{X\,(g) + Y\,(g) \rightarrow Z\,(g) + T\,(g)} \qquad (4)$$

and a similar vessel:



One could try and say that in this vessel, these two reactions occur:

$$\mathrm{X\,(g) + Y\,(g) \rightarrow Z\,(g) + T\,(g)} \qquad (5)$$
$$\mathrm{Z\,(g) + T\,(g) \rightarrow X\,(g) + Y\,(g)} \qquad (6)$$

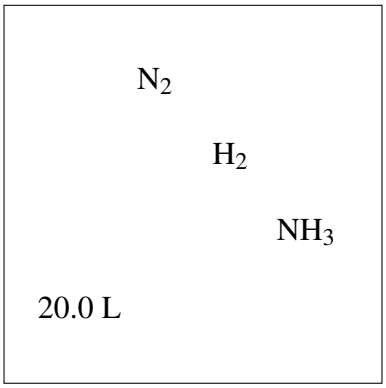
The difference here is that, whilst the rate of reaction (5) would be reasonable, the rate of reaction (6), the so-called "back reaction", would be **extremely slow** so as to be negligible, no matter what the concentrations of Z and T are. In fact, you'd even say that reaction (6) would not occur at all, so slow is the rate.

This means that, effectively, reaction (4) is an **irreversible** reaction. It (essentially) only occurs in the forward direction. In fact, this reaction would be considered **spontaneous**. The equilibrium constant of this sort of reaction would be extraordinarily high[116].

[116] All reactions can be considered reversible, technically, otherwise the K value of some reactions would be infinity. It's just that some reactions have K values as high as 10^{10} which means, that these are, effectively, irreversible reactions as the concentrations of the reactants must be infinitesimally small.

Question 2 (b) [Go to **Model Solution**]

Let us gather the information that we have.



The equilibrium constant (K) is 1.18×10^{-1}.

Let's review the reaction equation:

$$N_2\ (g) + 3H_2\ (g) \rightleftharpoons 2NH_3\ (g)$$

From this, we can write down the expression for the equilibrium constant:

$$K = \frac{[NH_3]_{eq}^2}{[N_2]_{eq}[H_2]_{eq}^3} = 1.18 \times 10^{-1}$$

The important thing here is that these concentrations are **equilibrium concentrations**, the concentrations of the reactants and products when the system is in equilibrium.

We know a quantity of N_2 and a quantity of H_2 and are asked to find the quantity of NH_3 at equilibrium. Clearly, the above equation represents the relationship between the quantities of these three gases. The problem here is that the question gave the quantities of the two gases as amounts (in mol), and the relationship above is given in **concentrations**. Therefore, we will need to convert the amounts of the given gases into concentrations.

From the question stem:

> Consider a 20.0 L rigid vessel containing a mixture of all three gases **in equilibrium**.

you can tell that the amounts of gases given here are those in equilibrium. Determining their corresponding concentrations knowing the volume of the vessel is 20.0 L:

$$[N_2]_{eq} = \frac{1.05}{20.0}$$

⇐ The amount of nitrogen gas in the vessel at equilibrium is 1.05 mol, and $c = \frac{n}{V}$.

$$= 0.0525 \text{ mol L}^{-1}$$

$$[H_2]_{eq} = \frac{3.24}{20.0}$$

⇐ The amount of hydrogen gas in the vessel at equilibrium is 3.24 mol, and $c = \frac{n}{V}$.

$$= 0.162 \text{ mol L}^{-1}$$

Now that we know the concentrations of N_2 and H_2 at equilibrium, we can use the equilibrium expression above to find the concentration of NH_3 and hence its amount:

$$
\begin{aligned}
K &= \frac{[NH_3]^2_{eq}}{[N_2]_{eq}[H_2]^3_{eq}} \\
[NH_3]^2 &= K[N_2][H_2]^3 && \Leftarrow \text{Rearranged the equation to make } [NH_3] \text{ the subject.} \\
&= 0.118 \times 0.0525 \times 0.162^3 && \Leftarrow \text{Remember that } K = 1.18 \times 10^{-1} = 0.118. \\
&= 2.63 \times 10^{-5} \\
[NH_3] &= 5.13 \times 10^{-3} \text{ mol L}^{-1} \\
n(NH_3) &= 5.13 \times 10^{-3} \times 20.0 && \Leftarrow \text{Remember the volume is 20.0 L in the vessel, and } n = cV. \\
&= 0.103 \text{ mol}
\end{aligned}
$$

Question 2 (c) (i) [Go to **Model Solution**]

Let us look at the reaction equation again:

$$N_2\ (g) + 3H_2\ (g) \rightleftharpoons 2NH_3\ (g)$$

We are asked to "explain the impact of the following on the **yield of ammonia produced**". Here, the yield of product in a reversible reaction is just code for **how far to the right the equilibrium position goes**.

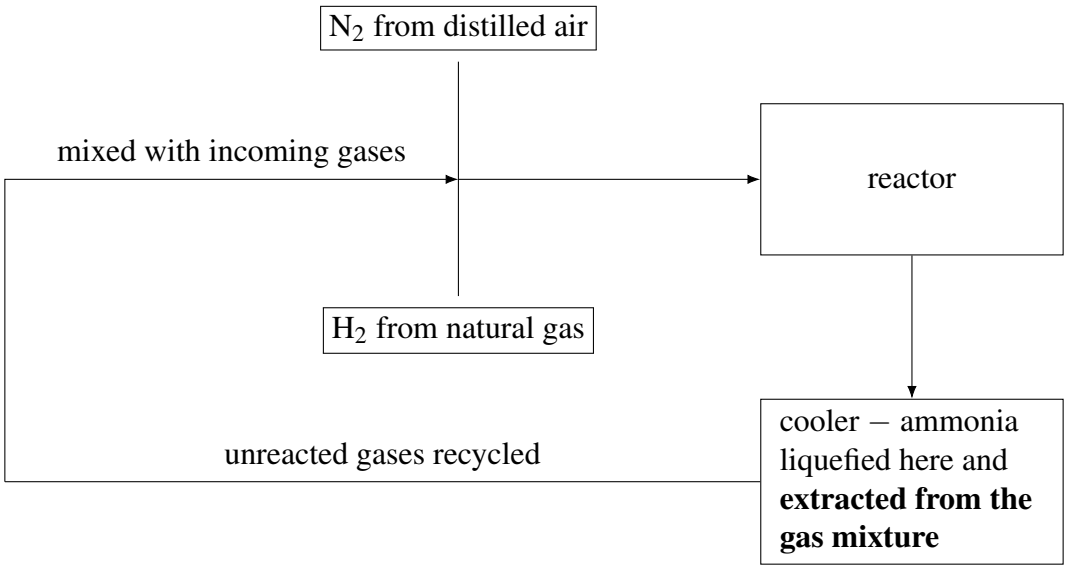
Remember!
The percentage yield of a product in a reaction is the percentage of the **maximum** possible amount of product that can be produced from the amounts of the given reactants, assuming the reaction goes to completion.
For example, suppose that, for the reaction

$$Mg\ (s) + 2HCl\ (aq) \rightarrow MgCl_2\ (aq) + H_2\ (g)$$

we mix 1 mole of Mg with 2 moles of HCl in a beaker and collect the H_2 and say we manage to collect 0.8 moles of H_2, then we can say that:

- The maximum possible amount of H_2 that can be produced here would be 1 mole (based on simple stoichiometry).
- The amount of H_2 produced in this experiment is actually 0.8 moles.
- The percentage yield would be 80%.

In other words, we need to determine how each action affects the equilibrium position.



In the cooler, ammonia (NH_3), a product, is "extracted from the gas mixture". In other words, we are **removing product** from the equilibrium mixture. How does this affect equilibrium position? Don't forget Le Chatelier's Principle:

Le Chatelier's Principle.
If an equilibrium system

- is subject to a **change**,
- the system will act as to **partially oppose** the change.

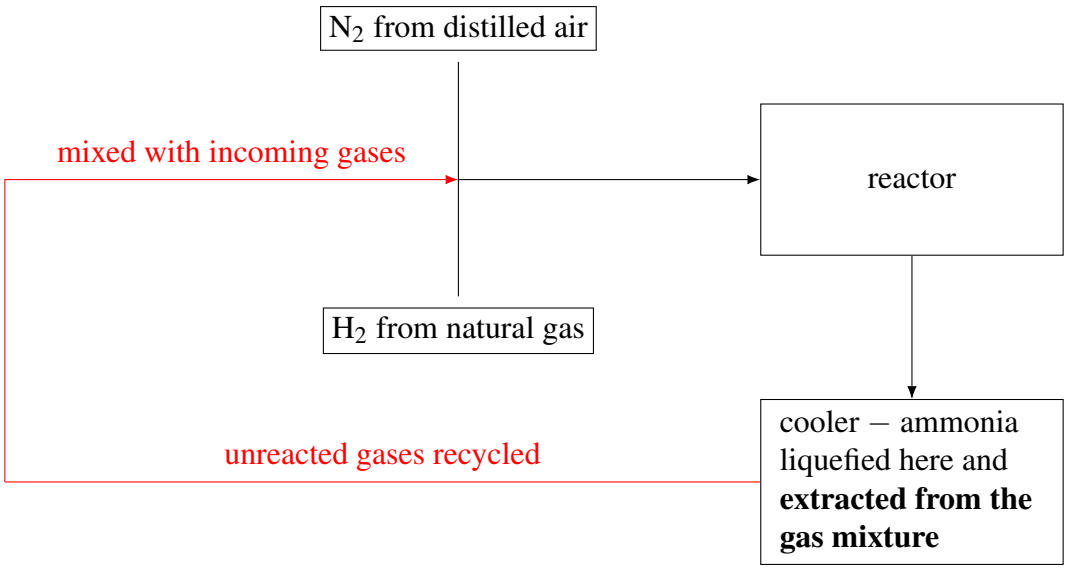
This is applicable **ONLY** when the following parameters are changed:

- **temperature**
- **concentration of a SINGLE reactant/product** (by addition or removal of a single reactant or product)
- **volume of the system.**

Here, we are removing product (NH_3). This means that the concentration of NH_3, a product, has decreased. The system will oppose this decrease by favouring the reaction that would lead to the subsequent slight increase (from the previously decreased value) in [NH_3]. This would be the forward reaction, where NH_3 is produced. In other words, **the equilibrium position would move right**. There would be a lower amount of reactants overall, since more of them were converted to products. There would be a higher **total** amount of NH_3, if we include the NH_3 that was extracted from the cooler and removed from the equilibrium system. So, the yield of NH_3 will be higher as a result of the actions of the cooler.

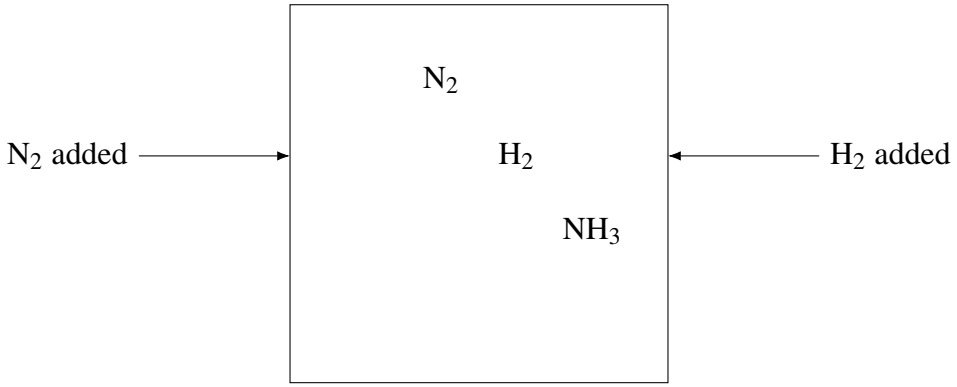
Question 2 (c) (ii) [Go to **Model Solution**]

The recycled gases are here:



Here, the recycled gases are being mixed with fresh N_2 and H_2 gases. The recycled gases are an equilibrium gas mixture containing N_2, H_2 and NH_3 (a small amount of NH_3 will have formed from N_2 and H_2 in the gas mixture that comes out of the cooler as the gases re-establish equilibrium after the cooler removes the NH_3 originally).

So, we have this situation:



In other words, we are **adding reactants** to an equilibrium gas mixture.

Again, via Le Chatelier's Principle, the system will favour the reaction that opposes what we just did (increasing reactant concentrations), by consuming reactants and hence decreasing their concentration. To have a net consumption of reactants, we need to favour the forward reaction.

In other words, the yield of ammonia increases.

Question 2 (c) (iii) [Go to **Model Solution**]

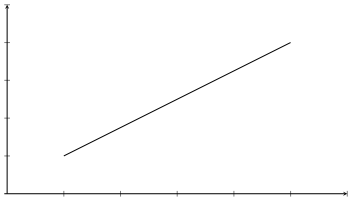
> **Remember!**
> In a reversible reaction, catalysts do **not** change equilibrium position. They increase the rate of both the forward and back reactions equally - so that whilst there is more rapid interconversion of reactants and products, the amount of reactant and product remain unchanged.

The use of a metal catalyst in the **reactor**, where the three gases are allowed to establish equilibrium, would not change the equilibrium position and hence would not affect the yield of ammonia produced. It would increase the rate at which equilibrium is established, meaning that the gases could be processed through the reactor in less time!

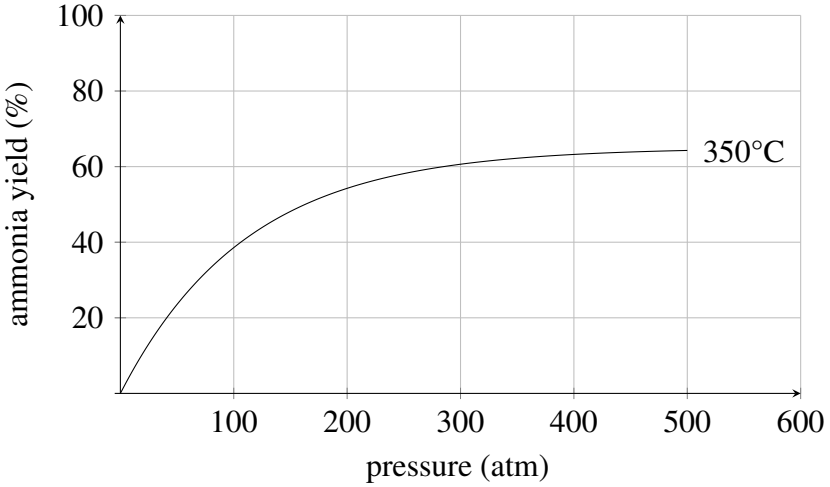
Question 2 (d) [Go to **Model Solution**]

> ...the ammonia yield-pressure characteristic...for each temperature has a **positive slope**.

A positive slope means that as you go from left to right on the graph, the curve is moving "upwards", like this:



Let's look at the real graph - isolating, say the 350°C graph:



As you can see, as we move to the right (**increasing pressure**) in the graph, the further up the curve goes (**increasing ammonia yield**). In other words, as the pressure in the vessel increases, the yield of ammonia increases too[117]. We are expected to explain why this happens, using Le Chatelier's Principle.

[117] Given that the temperature is kept constant here, we can only assume that the higher pressure is caused by increasing the concentrations of all reactants/products equally. In reality, when performing the reaction at higher pressure, we would simply do so by adding a larger amount of gases into the reaction vessel

Le Chatelier's Principle.
If an equilibrium system

- is subject to a **change**,
- the system will act as to **partially oppose** the change.

This is applicable **ONLY** when the following parameters are changed:

- **temperature**
- **concentration of a SINGLE reactant/product** (by addition or removal of a single reactant or product)
- **volume of the system**
- **pressure of the system by changing the concentrations of all reactants and products equally, or by changing the volume of the system.**

Another reminder above.

Let's use Le Chatelier's Principle here. We are increasing the pressure of the system[118]. The system would favour the reaction that decreases the pressure in the system - that is, the reaction that produces **fewer** gas molecules.

Looking at the reaction equation:

$$N_2\ (g) + 3H_2\ (g) \rightleftharpoons 2NH_3\ (g)$$

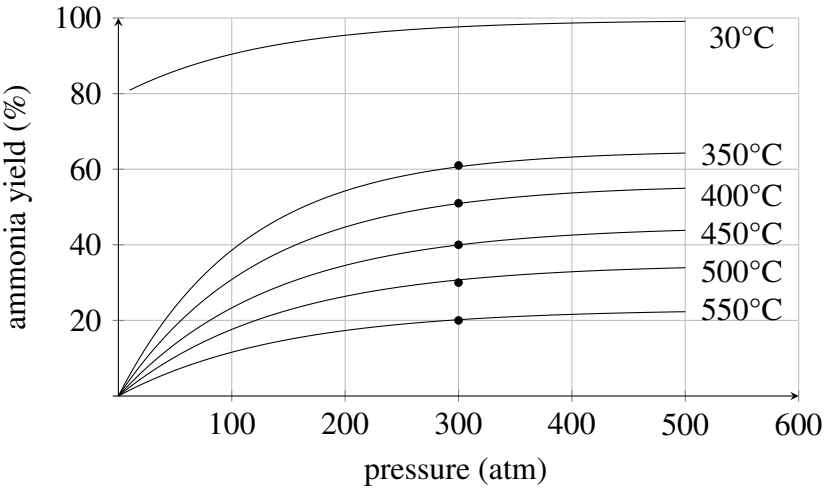
the LHS of the equation has $1 + 3 = 4$ gas molecules, whereas the RHS of the equation has 2 gas molecules. Hence, going from LHS to RHS would convert 4 molecules to 2 molecules. So, favouring the forward reaction would lead you to produce fewer gas molecules. Hence, the forward reaction is favoured if the pressure in the system were to increase for this reaction. This leads to an increase in the ammonia yield, which is what we see in the graph.

[118]Le Chatelier's Principle would only apply to pressure changes, if the pressure change occurred by changing the volume in the system (whilst keeping the same amount of reactant and product molecules) or by changing the concentration of each reactant and product by the same amount.

Question 2 (e) [Go to **Model Solution**]

How can the information given in the question stem determine information regarding the enthalpy of reaction here? Remember that, for all intents and purposes, the enthalpy of reaction is the change in chemical energy that occurs in the reaction. It is the ΔH value you would have learnt about earlier in the year.

Looking through the information provided in the question stem, only one piece of information would give you a clue as to what the enthalpy of reaction could be:



Here, I have labelled coordinates denoting the yield of ammonia at 300 atm of pressure at various temperatures. As you can see, the higher the temperature going from 350°C to 550°C, the **lower** the yield, going from about 60% down to about 20%.

This should remind you of this:

Remember!
When you **increase the temperature** (thermal energy) of a system, the system will favour the reaction that consumes thermal energy:

- if the forward reaction is exothermic, the back reaction would be favoured (as it is the endothermic reaction)
- if the forward reaction is endothermic, the forward reaction would be favoured (as it is the endothermic reaction)

Question 2 (f) [Go to **Model Solution**]

This is a common question you will be asked - you will be given a reversible reaction that is used to produce a chemical in industry, and you will be asked to explain why a particular set of reaction conditions are used.

The commonsense principle here is that you want to produce the chemical:

- with the highest possible yield - so as to minimise the amount of starting material wasted
- as fast as possible - because this means earning money $$$
- at the lowest cost possible.

Yield is dependent on equilibrium position - you want to, as much as possible, push the equilibrium position to the right.

The reaction rate should be as fast as possible to produce the chemical as fast as possible[119].

In reality, the three goals listed above are at odds with one another. A measure to increase reaction rate, such as increasing the temperature of the reaction vessel, would increase costs (as it requires more energy to keep the reaction vessel hotter), and could also decrease yield in certain circumstances[120]. In such situations, you will end up compromising and finding the best possible rate/yield/cost combination that will give you the best possible outcome[121].

On this occasion, we have selected a temperature and a pressure. Let's see how these two would affect the rate and yield of NH_3 produced:

- **Temperature**

  The higher the temperature, the higher the rate at which equilibrium is established and, ultimately, the higher the rate of NH_3 produced[122]. So, theoretically, you can have as high a temperature as you want, and you'll get a faster reaction rate. This is constrained, ultimately, by cost.

  The higher the temperature, as shown in the graph in the question stem, the **lower** the yield of NH_3 - from which we had previously extrapolated that the reaction must be exothermic.

  This means that higher temperatures mean a higher rate of reaction (good), but lower yield of NH_3 (bad) and higher costs (bad).

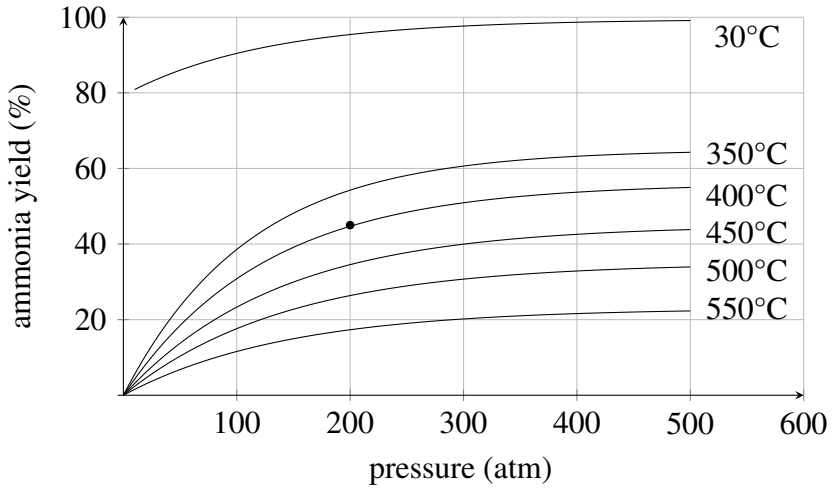
[119] For reversible reactions, what we really mean is that equilibrium should be established as fast as possible - which would require the forward and back reactions to both be rather fast.

[120] Can you guess what the circumstances are? Don't look yet. OK, you can look. The yield would decrease with increasing temperature if the reaction were exothermic.

[121] In other words, as much $$$ as possible, as fast as possible.

[122] Think about it. If we have a high rate at which equilibrium is established, the gas mixtures will not have to spend nearly as much time in the reactor, so gases can cycle between the reactor and cooler at a high rate, with NH_3 being extracted faster as the gases will be transferred to the cooler at a higher rate (since they don't have to spend as much time in the reactor.

So, we use a moderately high temperature of 400°C because this would presumably give rise to a sufficiently rapid equilibrium establishment (i.e. higher reaction rate) and hence a more rapid production of NH_3, without compromising too much on the yield of NH_3:



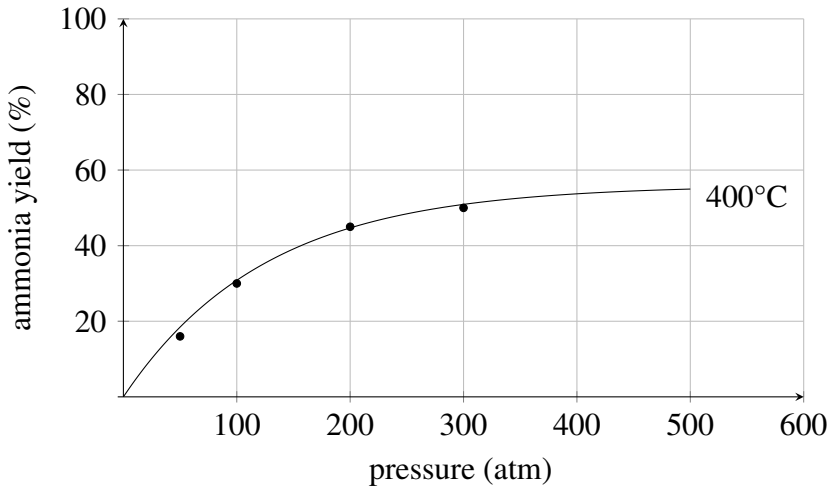
As you can see, the yield of NH_3 is about 45%. Not too bad. As you can see, the yield drops further if we increase the temperature from 400 to 550°C - down to less than 20%, which isn't great. And for what? The reaction rate probably would not increase by enough to justify a plummeting of the yield.

- **Pressure**

  The higher the pressure[123], the higher the reaction rate (obviously!) and, as described in a previous question, the higher the yield of ammonia. So, win-win right? Just increase the pressure to over 9,000 atm right? Maybe not. That would be extremely expensive.

  We would thus choose a pressure which is essentially cost-effective.

  Looking at the provided diagram, we can see why:



You can see that I selected a few points in the graph depicting the yield-pressure relationship at 400°C:

[123]Remember the caveats we discussed earlier. We really mean the higher the pressure as a result of an increase in the concentrations of all reactants and products by the same fraction.

- A pressure of 1 atm, at a hot room temperature (30°C), would give you a yield of just over 80%. Problem is, making the reaction vessel hot to 400°C whilst maintaining atmospheric pressure would give you a yield just over 0%. From 80% to just over 0%. Better be worth it. Not at atmospheric pressure.
- What about at 200 atm? The yield goes from 0% to about about 45%. Not bad at all. AND you don't have to wait forever for equilibrium to be established because you can use 400°C as the temperature. Even better, increased pressures[124] cause higher reaction rates in their own right, on top of increasing temperature!

  Now, it'll be a little expensive and energy-intensive, pumping gases in at 200 atm rather than atmospheric pressure, but for a yield of 45% and very rapid establishment of equilibrium, it's actually pretty cost effective.
- Why not increase the pressure further, say to 300 atm you ask? The yield would increase further AND the rate of reaction would increase further. But look - the yield would not increase by much, only to about 50% (from 45%). Presumably it would be way too expensive (due to the energy required to pump gases at 300 atm and you'd have to use more expensive equipment that can withstand such pressures), and you wouldn't get much additional benefit. The important thing here is not just that you do get a benefit - you have to have an idea as to **how much** benefit you get, particularly when costs become an issue.

So, we have our reasons for using the given temperature and pressure.

Question 2 (g) [Go to **Model Solution**]

At first, this question would not make sense - we know that catalysts do not change equilibrium position, so how can they affect the yield of ammonia here? The key word that gives you the clue is "indirectly".

Have a look at the previous Detailed Solution - particularly the reference to performing the reaction at 30°C.

Performing the reaction at atmospheric pressure and 30°C temperature, you'd get a yield of about 80%, way higher than the 45% that you settle for in the previous question. Hell, if you use 200 atm in pressure, you could get an even better yield, up to 95%!

The reason why, ordinarily, we cannot use a temperature of 30°C is that presumably the reaction would be so slow that it would take forever for equilibrium to be established among the N_2, H_2 and NH_3 gases that the rate of NH_3 production and extraction in the cooler would be so slow. Shame, because the yield would have been amazing!

Now, if we had an enzymatic catalyst which dramatically increases the reaction rate (beyond the effects of high temperature and pressure), that barrier is gone! So, we would have a high reaction rate, AND amazing yields! In other words, using an enzymatic catalyst means that it is possible to get higher yields whilst maintaining a sufficient reaction rate to be commercially viable. Also, it would be cheaper because you don't have to keep the reaction vessel hot, nor at high pressure[125].

[124] That is, by increasing concentration of reactants/products proportionally.

[125] Actually, it would probably not be that much cheaper. An enzymatic catalyst that gives a company that much extra value? The company that does produce such a catalyst would surely charge an arm and a leg and make humongous profits.

Question 3 (a) [Go to **Model Solution**]

Firstly we have to figure out which reaction occurs at the positive electrode. This cell is a **primary cell**, which is a type of **galvanic cell**.

Remember!
The polarities of the electrodes in a galvanic and electrolytic cell are shown below:

	anode (oxidation)	cathode (reduction)
galvanic cell	−	+
electrolytic cell	+	−

Now, the two reactions are:

- **Zn is converted to ZnO.**

  The oxidation number of Zn in Zn metal is 0, since Zn is an element.

  In ZnO, since this is an ionic compound composed of Zn^{2+} and O^{2-} ions, the oxidation number of Zn here is **+2**.

  Since the oxidation number is increasing here, this is an **oxidation** reaction which occurs at the anode.

- **MnO_2 is converted to Mn_2O_3.**

  The oxidation number of Mn in MnO_2 is **+4**, since the oxidation number of O is -2 (and the sum of oxidation numbers of the Mn and each of the two O atoms has to be 0).

  Similarly, the oxidation number of Mn in Mn_2O_3 is **+3**, as the total sum of oxidation numbers of the three O atoms is -6; this means that sum of oxidation numbers of the two Mn atoms must be +6, so each Mn atom must be +3.

  Since the oxidation number is decreasing here, this is a **reduction** reaction that occurs at the cathode.

So, MnO_2 is reduced to Mn_2O_3 at the cathode. Now, we have to write a half-equation for this reaction in **base**. How do we know that it is in base? It is stated that the electrolyte is a 7 M KOH paste, which is a base. Therefore, we can use either the KOHES (base) or the KOHEBS method.

Balancing half-equations in base

The two methods by which you could balance a redox half-equation in base are below; here, we have to assume that there are ample amounts of OH^- ions in solution, and hardly any H^+, so we cannot use H^+ ions to balance the reaction half-equation, but we can use OH^-.

The KOHES (base) method is below:

1. Balance **K**ey atoms.
2. Balance **O** atoms by adding H_2O molecules to the appropriate side of the reaction equation.
3. Balance **H** atoms by adding H_2O / OH^- "pairs" on either side of the reaction equation[a].
4. Balance charge by adding **E**lectrons to the appropriate side of the reaction equation.
5. Add **S**tates.

The KOHEBS method is below; here, you pretend you are balancing the reaction equation in acid so you can use H^+ ions, then you do a little trick at the end to remove the H^+ ions and add OH^- ions:

1. Balance **K**ey atoms.
2. Balance **O** atoms by adding H_2O molecules to the appropriate side of the reaction equation.
3. Balance **H** atoms by adding H^+ ions
4. Balance charge by adding **E**lectrons to the appropriate side of the reaction equation.
5. Convert the reaction equation in acid written to a reaction equation in **B**ase by adding the same number of OH^- ions on either side of the reaction equation, so that it cancels out the H^+, converting it to water[b].
6. Add **S**tates.

[a] When I say H_2O / OH^- "pairs", I mean adding a H_2O on one side of the equation, and an OH^- to the other side of the equation. This would have the net effect of adding 1 excess H atom on the side of the equation in which you added the H_2O molecule.

[b] H^+ (or rather, H_3O^+) cannot coexist in significant quantities with OH^- in solution; they'd just react with one another to form H_2O.

Let's use the KOHEBS method here:

1. Balance **K**ey atoms.

   $$MnO_2 \rightarrow Mn_2O_3$$

   There is 1 Mn atom on the LHS and 2 Mn atoms on the RHS. Therefore, we will add another MnO_2 to the LHS to balance out the Mn atoms:

   $$2\ MnO_2 \rightarrow Mn_2O_3$$

2. Balance **O** atoms by adding H_2O molecules to the appropriate side of the reaction equation.

There are 4 O atoms on the LHS but 3 on the RHS, hence a H_2O molecule must be added to the RHS to balance out the O atoms:

$$2\ MnO_2 \rightarrow Mn_2O_3 + H_2O$$

3. Balance **H** atoms by adding H^+ ions (pretending we are balancing in acid).

   There are no H atoms on the LHS and 2 on the RHS, hence we add 2 H^+ ions to the LHS to balance this out:

$$2\ MnO_2 + 2\ H^+ \rightarrow Mn_2O_3 + H_2O$$

4. Balance charge by adding **E**lectrons to the appropriate side of the reaction equation.

   The charge on the LHS is +2, the charge on the RHS is 0. Therefore, 2 electrons must be added to the LHS to reduce the charge on the LHS to 0, equal to that of the RHS:

$$2\ MnO_2 + 2\ H^+ + 2\ e^- \rightarrow Mn_2O_3 + H_2O$$

5. Convert the reaction equation in acid written to a reaction equation in **B**ase by adding the same number of OH^- ions on either side of the reaction equation, so that it cancels out the H^+, converting it to water.

   Remember that the reaction actually occurs in base, so there are OH^- ions around, not H^+ ions. A method we could use to convert a reaction half-equation in acid to one in base is add 2 OH^- ions to each side of the reaction half-equation, knowing that H^+ and OH^- would react to form water:

$$2\ MnO_2 + 2\ H^+ + 2\ e^- \rightarrow Mn_2O_3 + H_2O$$

$$2\ MnO_2 + 2\ H^+ + 2\ OH^- + 2\ e^- \rightarrow Mn_2O_3 + H_2O + 2\ OH^- \quad \Leftarrow \text{Adding 2 } OH^- \text{ ions to each side, to get the same number of } H^+ \text{ and } OH^- \text{ ions on one side of the equation that can be cancelled out.}$$

$$2\ MnO_2 + 2\ H_2O + 2\ e^- \rightarrow Mn_2O_3 + H_2O + 2\ OH^- \quad \Leftarrow \text{On the LHS, the 2 } OH^- \text{ and 2 } H^+ \text{ can be replaced by 2 } H_2O\text{, as } H^+ \text{ and } OH^- \text{ cannot coexist in significant amounts, and would just react with each other to form water.}$$

$$2\ MnO_2 + H_2O + 2\ e^- \rightarrow Mn_2O_3 + 2\ OH^- \quad \Leftarrow 2\ H_2O \text{ on the LHS, 1 } H_2O \text{ on the RHS. We can cancel out 1 } H_2O \text{ from each side of the equation; there will be 1 } H_2O \text{ remaining on the LHS, and none on the RHS.}$$

6. Add **S**tates.

$$2\ MnO_2\ (s) + H_2O\ (l) + 2e^- \rightarrow Mn_2O_3\ (s) + 2\ OH^-\ (aq)$$

Question 3 (b) [Go to **Model Solution**]

This is a galvanic cell, where chemical energy from a redox reaction is utilised to drive an electrical current. The energy transformation here is from **chemical to electrical**.

Question 3 (c) [Go to **Model Solution**]

Remember!
The flow of ions inside the electrolyte/salt bridge will always attempt to balance the polarisation in the cell caused by a mass transfer of electrons (and negative charge) from one half cell to the other half cell.
In particular, given electrons move towards the cathode (in any cell) as the cell operates, causing a slight buildup of negative charge at the cathode momentarily:

- cations will move towards the cathode, as they are positively charged and attracted to the extra negative charge [a] arriving at the cathode
- anions will move towards and anode, as they are negatively charged and attracted to the relative positive charge[b]

[a] This is distinct from the **polarity** of the electrode. The cathode has a positive polarity in a galvanic cell, and a negative polarity in an electrolytic cell. However, there will always be a momentary net negative charge at the cathode no matter what cell type we have.
[b] Again, this is distinct from the **polarity** of the electrode.

Since K^+ is positively charged, it will move towards the part of the cell that is accumulating electrons (i.e. the cathode). Hence, the K^+ ions will be moving **away** from the Zn electrode and onto the other electrode (the top outer casing).

Question 4 (a) (i) [Go to **Model Solution**]

Let us break down the name.

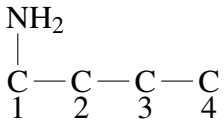
3-chlorobutan-1-amine

Main carbon skeleton (click here)

- "but" means a 4-carbon skeleton
- "an" means no carbon-to-carbon double or triple bonds
- "-1-amine" means that there is an amino ($-NH_2$) group[126] on carbon-1 (click here)

[126] Remember that, when numbering carbon atoms, the **principal functional group** (the functional group that goes at the end of the molecule name) dictates the carbon numbering. You number the carbon atoms in the direction that gives you the lowest possible number on the principal functional group. If there is no principal functional group, then you do the same thing for any C=C double bonds. If there are no principal functional groups or C=C double bonds, you would do the same thing for the substituents.

which gives you a skeleton that looks like this so far:

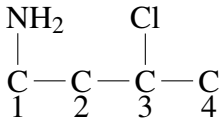


Substituents

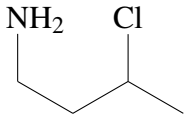
We have the phrase "3-chloro".

This means that there is a -Cl group on carbon-3.

Placing this −Cl functional group on the carbon-3 atom like so:



Now we have allocated the substituents, we can proceed to fill in the H atoms:



Question 4 (a) (ii) [Go to **Model Solution**]

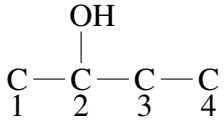
Similarly, we can do the same thing here:

3-methylbutan-2-ol

Main carbon skeleton (click here)

- “but” means a 4-carbon skeleton
- “an” means no carbon-to-carbon double or triple bonds
- “-2-ol” means that there is a hydroxyl (-OH) group on carbon-2 (click here)

which gives you a skeleton that looks like this so far:



Substituents

We have the phrase "3-methyl".

This means that there is a -CH_3 group on carbon-3.

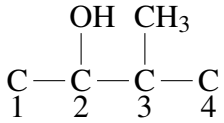
Revision!

Substituents are named as below:

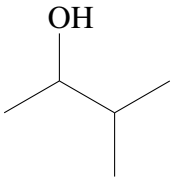
chloro-	$-Cl$
bromo-	$-Br$
iodo-	$-I$
methyl-	$-CH_3$
ethyl-	$-CH_2CH_3$
amino-[a]	$-NH_2$

[a] Amino ($-NH_2$) groups are usually principal functional groups whose affix is usually stuck at the end of the name (as -amine). These groups only become substituents when the molecule also has a hydroxyl ($-OH$) group or a carboxyl group ($-COOH$).

Placing this $-CH_3$ functional group on the carbon-3 atom like so:

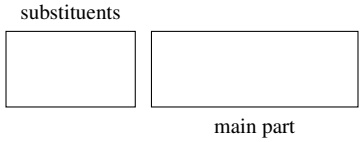


Now we have allocated the substituents, we can proceed to fill in the H atoms:

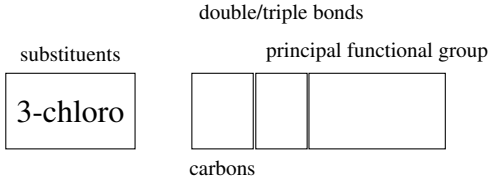


Question 4 (b) (i) [Go to **Model Solution**]

Let us recall the parts of the IUPAC name:



Further dividing this:



When determining the IUPAC name of a compound from its structure, take the following approach:

1. **Establish the "main part" of the molecule - including the carbon skeleton**.

   The main part of the molecule will contain the **longest possible carbon chain** that contains a principal functional group (either the $-OH$, $-NH_2$ or $-COOH$ group) and/or a C=C double bond (or triple bond), if these are present.

   From these, you can find the number of C atoms in the chain, and establish the presence of C=C bonds and the principal functional group (if any) and, then, determine the name of the "main part" of the molecule.

2. **Identify the substituents and determine their location in the carbon chain (numbering the C atoms in the skeleton)**.

   When numbering the C atoms in the carbon skeleton, number the carbon atoms in the direction that:

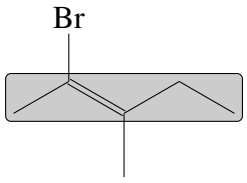
   - gives the principal functional group the **lower number** if it is present;
   - gives the C atom in a C=C double bond the lower number (if there is NO principal functional group but there is a C=C double bond);
   - gives the sum of the carbon numbers of all substituents the lower value (if there is no principal functional group or C=C double bond); for example, 1,1,1-trichloroethane rather than 2,2,2-trichloroethane is the appropriate name for the compound with semistructural formula $CHCl_3CH_3$

   Then, write the substituents in **alphabetical order** (of the substituents, not any "di" or "tri" prefixes), NOT in numbering order. For example, "4-chloro-3-ethyl" rather than "3-ethyl-4-chloro".

So, let's go through each step.

1. **Establish the "main part".**

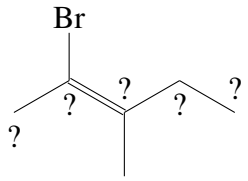
   This is easy enough - the longest carbon chain is shown below:

   From here:

   - 5 carbon atoms – this gives you "pent"
   - a single C=C bond gives you "ene"
   - this C=C bond starts at carbon-2:

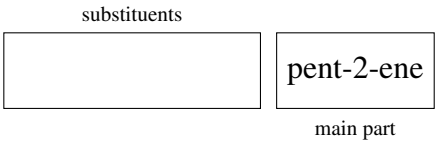
Remember that we could number from left to right here giving us carbon-2, or from right to left here giving us carbon-3. However, numbering from left to right gives you the lower number, so we should stick with this numbering here.



Hence, the affix we use is “-2-ene”.

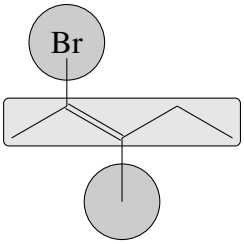
The main part of the molecule is, therefore, "pent-2-ene".

We are now at this stage:



2. **Identify the substituents and determine their location in the carbon chain (numbering the C atoms in the skeleton)**.

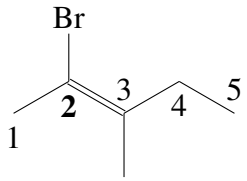
   Our substituents are circled below:



   We can recognise the substituents as:

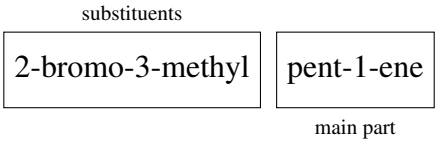
   - the CH_3- group has the affix "methyl", and there is only one of these groups
   - the $Br-$ group has the affix "bromo", and there is only one of these groups

Now, we have to state **where** these groups are in the carbon skeleton. We have already numbered the carbon atoms (the C=C double bond dictated the direction of numebring already).



- bromo on carbon-2 $\rightarrow$ 2-bromo
- methyl on carbon-3 $\rightarrow$ 3-methyl

x In the name, we list the substituents in alphabetical order (NOT numbering order), with the "b" in "bromo" coming before the "m" in "methyl". Hence, we get:

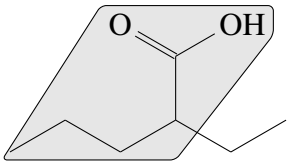


The answer, therefore, is 2-bromo-3-methylpent-1-ene.

Question 4 (b) (ii) [Go to **Model Solution**]

Rinse and repeat.

1. **Establish the "main part".**



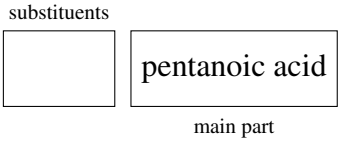
Trick here is that the principal functional group (-COOH) must be included in the chain, even if there is a possible carbon chain that is longer but does not include the principal functional group.

From here:

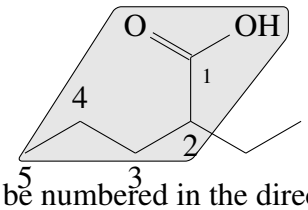
- 5 carbon atoms − this gives you "pent"
- no C=C bonds gives you "an(e)"
- the carboxyl (-COOH) group gives you “-oic acid”

You are left with “pentanoic acid”.

We are now at this stage:



2. **Identify the substituents and determine their location in the carbon chain (numbering the C atoms in the skeleton)**.



Remember that the carbon atoms must be numbered in the direction such that the principal functional group is associated with the lowest possible carbon number. Since the carboxyl group (the principal functional group) must be at the end of the molecule, its carbon atom **must** be carbon-1.

This gives you the above numbering.

As you can see, there is a CH_3CH_2 group on carbon-2. This is an **ethyl** group. Therefore, the affix used here will be "2-ethyl".

We end up with this:

substituents	main part
2-ethyl	pentanoic acid

The name of the molecule is 2-ethylpentanoic acid.

Question 5 (a) [Go to **Model Solution**]

Remember!
The polarities of the electrodes in a galvanic and electrolytic cell are shown below:

	anode (oxidation)	cathode (reduction)
galvanic cell	−	+
electrolytic cell	+	−

The **negative electrode** in an electrolytic setup is the **cathode**.

So, what reaction occurs at the cathode? Recall that, by definition, the cathode is the site of **reduction**. We need to have a look at the species that would get reduced. To do this, let us list the species present in solution:

$$Zn^{2+}\ (aq) \qquad SO_4{}^{2-}\ (aq) \qquad H_2O\ (l)$$

Now, the strongest oxidant - the species that has the greatest tendency to accept electrons at the cathode - will be reduced at the cathode[127].

The question is - which is the strongest oxidant?

[127]The two exceptions you will come across in the course are $SO_4{}^{2-}$ and $NO_3{}^-$ ions. These **never** get discharged at the cathode. Why? Problem is that they are **negatively charged** ions. Remember how earlier we had discussed how anions congregate towards the anode, where there are momentary positive charges borne out by the removal of electrons from the anode? Well, if anions are congregrating towards the anode, they cannot exactly be discharged at the cathode.

Since oxidants get **reduced** and the half-equations in the electrochemical series are reduction equations, let us list the equations involving either Zn^{2+} or H_2O on the **left** hand side in the electochemical series:

$$Zn^{2+}\ (aq) + 2e^- \leftrightharpoons Zn\ (s) \qquad E^\circ = -0.76\ V$$
$$2H_2O\ (l) + 2e^- \leftrightharpoons H_2\ (g) + 2OH^-\ (aq) \qquad E^\circ = -0.83\ V$$

Remember, the greater the E° value (or less negative), the greater the tendency for the reaction to go forward. Hence, Zn^{2+} is the better oxidant and will therefore be reduced at the negative electrode the cathode. The reaction equation is:

$$Zn^{2+}\ (aq) + 2e^- \rightarrow Zn\ (s)$$

Question 5 (b) [Go to **Model Solution**]

Where does the change in mass of the anode come from? Note that the anode is made out of copper. Metals are rather good at giving away their electrons (with the exception of gold). Hence, the change in mass of the anode would be due to the oxidation of Cu to Cu^{2+}.

Now, we are practically asked to determine the mass of copper that is oxidised in 1 second. We know:

- the current through the cell
- the potential difference across the electrodes

We can determine the mass or amount of copper oxidised through determining the amount of electrons that leaves the anode. We are given information regarding this, namely that 1.00 C of charge leaves the anode in 1 second. We also know that 1 mol of electrons has a charge of 96500 C. Since only 1.00 C of charge left the anode:

$$\begin{aligned} n(e^-) &= \frac{1.00}{96500} \\ &= 1.04 \times 10^{-5}\ \text{mol} \end{aligned}$$

Now, for every 1 mole of Cu oxidised to Cu^{2+}, 2 moles of electrons are released. Since only 1.04 *times* 10^{-5} mol of e^- is released, we can say:

$$\begin{aligned} n(Cu) &= \frac{1.04 \times 10^{-5}}{2} \\ &= 5.18 \times 10^{-6}\ \text{mol} \end{aligned}$$

Hence, the mass of Cu oxidised is:

$$\begin{aligned} m(Cu) &= 5.18 \times 10^{-6} \times 63.5 \\ &= 3.29 \times 10^{-4}\ \text{g} \end{aligned}$$

Thus, the mass of the anode decreases by 3.29×10^{-4} g per second.

Question 5 (c) [Go to **Model Solution**]

When you turn up the current too high, this allows potential reductants present in the solution **other than the strongest reductant** to be oxidised at the anode[128].

What are the species at the anode that can potentially be oxidised?

- Zn^{2+} - this species **cannot** be oxidised any further, so definitely not this one.
- H_2O - this can be potentially oxidised to O_2
- Cu - remember that the anode is made out of Cu, and Cu can definitely lose electrons here!

Since we are talking about a high current here where multiple species can react at the anode, we can say that both Cu and H_2O can be oxidised at the anode here.

What are the reactions in this case? Read from the electrochemical series!

$$\text{Cu (s)} \rightarrow \text{Cu}^{2+}\text{(aq)} + 2\ \text{e}^-$$
$$2\ \text{H}_2\text{O (l)} \rightarrow \text{O}_2\ \text{(g)} + 4\ \text{H}^+\ \text{(aq)} + 4\ \text{e}^-$$

Question 5 (d) [Go to **Model Solution**]

Note that platinum is an inert[129] metal. This means that replacing the copper electrode with a platinum one **eliminates Cu as a possible reactant at the anode**.

Would this affect the composition of the coin? Well, let's see what happens normally (if we had a Cu electrode).

We can rightly figure out that Cu would be oxidised to form Cu^{2+} in the solution. This means that over time, you would have a significant amount of Cu^{2+} ions as well as Zn^{2+} ions. Well, that changes things, doesn't it? What would be very helpful is knowing in the back of your mind that **for a metal ion, Cu^{2+} is unusually good at accepting electrons** (Ag^+ and Au^+ fit this category too). This would raise the possibility that Cu^{2+} could compete with Zn^{2+} at the cathode.

A cursory look at the electrochemical series would help us notice that in fact Cu^{2+} is in fact better at accepting electrons than is Zn^{2+}. Hence, over time, Cu^{2+} would be increasingly preferentially discharged at the cathode, and hence the coin would end up being plated with a mixture of copper and zinc.

If you replaced the copper electrode with a platinum one, then there would be no Cu^{2+} released into the solution, only Zn^{2+} would be discharged at the cathode and the coin would be plated with zinc only.

[128] Technically speaking, this happens because the way you turn the current up in an electrolytic cell is by increasing the voltage across the power source and hence the voltage across the electrolytic cell electrodes. It's the increase in voltage that causes the side reactions to occur.

[129] Inert = unreactive. Or, if you watch the ancient but hilarious British comedy show *Yes, Minister*, inert means "not...ert". If you want to see the video clip I am referring to on YouTube - search "Explaining MetaDioxin Yes Minister" on YouTube and there is a 4 minute video of hilarious stuff on there, including a character's attempt to explain the term "inert". But I digress.

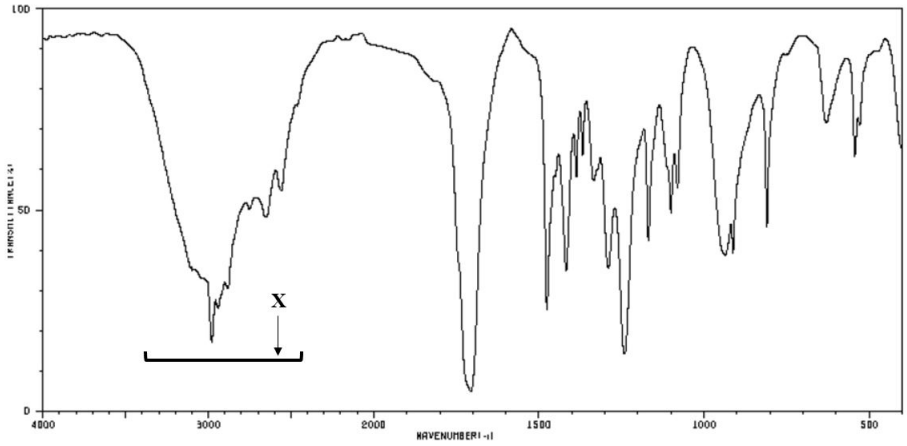
Question 6 (a) [Go to **Model Solution**]

> **Remember!**
> In a ^{1}H NMR spectrum, the **number of peaks**[a] is equal to the **number of hydrogen environments** in the molecule.
>
> [a] We are considering a split peak such as a triplet as a single peak.

A glance at the ^{1}H NMR spectrum shows that there are **3** peaks, and therefore **3** hydrogen environments.

Question 6 (b) [Go to **Model Solution**]

Have a good look at this absorption band labelled **X**.

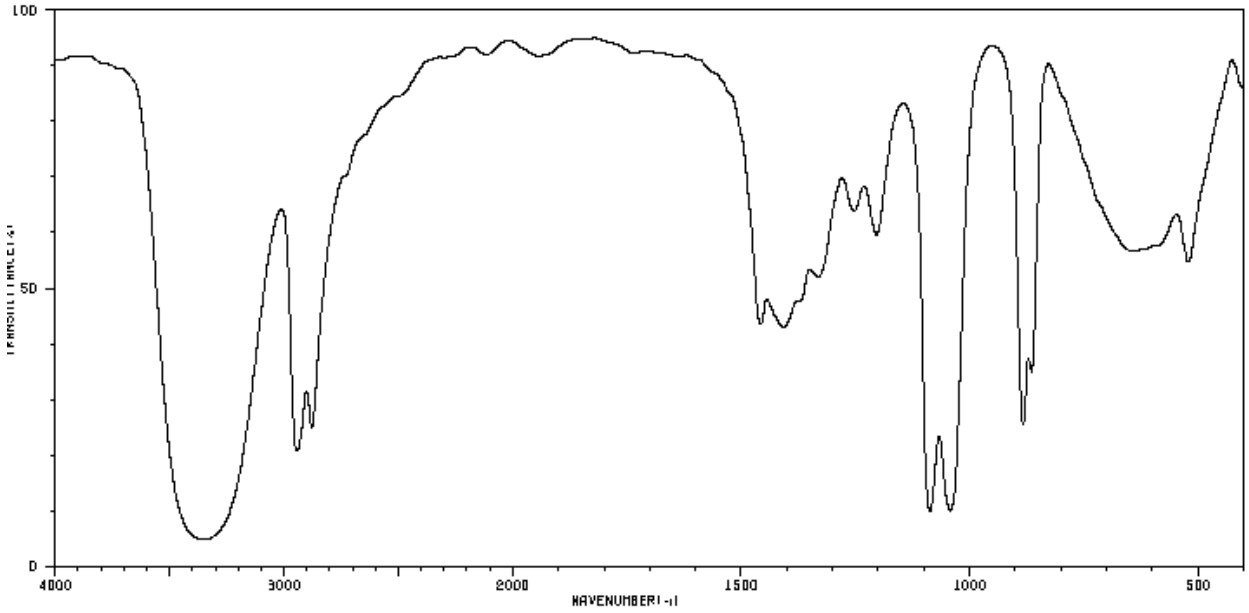


(Source: Spectral Database for Organic Compounds, National Institute of Advanced Industrial Science and Technology, Japan)

Notice how it looks like a **fat** and **broad** absorption band, with some **spiky** bits? Recognise this absorption band, because you are very likely to see it in your SACs and the exam. This is actually a superposition of two absorption bands - the O-H (acid) absorption band and the C-H absorption band. To see what the two peaks look like separately, let us compare the following absorption bands:

- C-H absorption band in a molecule of **ethane-1,2-diol** (at around 2900 cm^{-1})
- O-H (alcohol) absorption band in a molecule of **ethane-1,2-diol** (at around 3300-3400 cm^{-1})

IR spectrum of ethane-1,2-diol

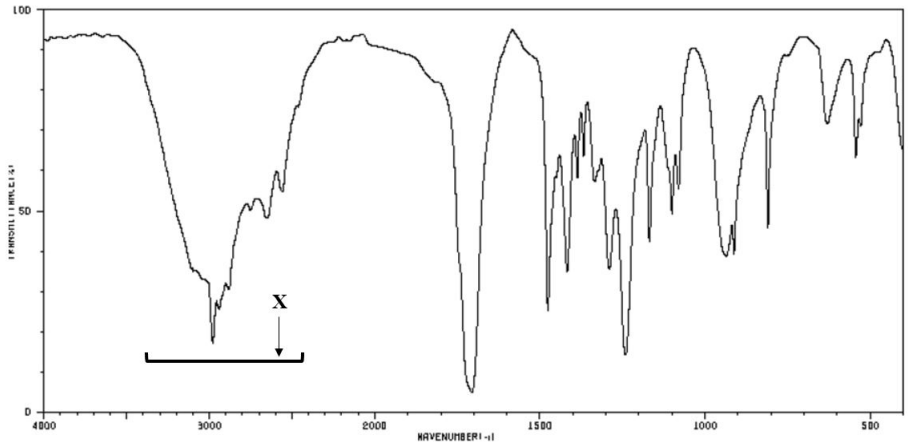


(Source: Spectral Database for Organic Compounds, National Institute of Advanced Industrial Science and Technology, Japan)

Note that:

- the C-H absorption band is “spiky”
- the O-H (alcohol) absorption band is “fat and broad”

Now, the only difference between an O-H (alcohol) and O-H (acid) absorption band is that the latter has a slightly lower wavenumber. If you imagine a molecule with a C-H absorption band (as all organic molecules have) and an O-H (acid) absorption band (such as a carboxylic acid), you will find that the two have very similar wavenumbers and will superimpose themselves on the IR spectrum, much like this:



(Source: Spectral Database for Organic Compounds, National Institute of Advanced Industrial Science and Technology, Japan)

Question 6 (c) [Go to **Model Solution**]

By definition, the **base peak** in a mass spectrum is the **highest peak** and the one designated a relative abundance of 100 (arbitrary units). In this mass spectrum, the m/z value (horizontal axis) of the base peak is at 43. Therefore, the answer is 43.

Question 6 (d) [Go to **Model Solution**]

Remember that the **empirical formula** is the smallest whole number ratio of each atom in a molecule.
We know that the molecule is 54.5% C, 9.1% H and 36.4% O by mass. A good approach to finding the empirical formula is assuming that we have **100 g** of the substance. If this is the case, then we have

$$\begin{aligned} m(\mathrm{C}) &= 54.5 \text{ g} \\ m(\mathrm{H}) &= 9.1 \text{ g} \\ m(\mathrm{O}) &= 36.4 \text{ g} \end{aligned}$$

To work out the empirical formula, we need to work out the **ratio** between the **amount (in mol)** of each of the atoms. Remember that, for example, a 1:2:3 mol ratio between the three atoms is equivalent to a 1:2:3 ratio of each atom, since the mole is a collection of atoms (like a dozen, or a score). Thus, we should work out the amount of each of the atoms, in mol:

$$\begin{aligned} n(\mathrm{C}) &= \frac{54.5}{12.0} = 4.54 \text{ mol} \\ n(\mathrm{H}) &= \frac{9.1}{1.0} = 9.1 \text{ mol} \\ n(\mathrm{O}) &= \frac{36.4}{16.0} = 2.28 \text{ mol} \end{aligned}$$

Now, to work out the whole number ratio between the amounts of each of the atoms, we divide by the smallest amount in mol (the 2.28 mol):

$$n(\mathrm{C}) : n(\mathrm{H}) : n(\mathrm{O}) = \frac{4.54}{2.28} : \frac{9.1}{2.28} : \frac{2.28}{2.28} = 2 : 4 : 1$$

Thus, the empirical formula is C_2H_4O.

Question 6 (e) [Go to **Model Solution**]

Given the empirical formula (C_2H_4O) has a molar mass of 44 g mol^{-1} ($12.0 \times 2 + 1.0 \times 4 + 16$), the molar mass of the molecule needs to be a whole number multiple of 44. The possible molecular formulae include:

- C_2H_4O (M = 44 g mol^{-1})
- $C_4H_8O_2$ (M = 88 g mol^{-1})
- $C_6H_{12}O_3$ (M = 132 g mol^{-1})

and so on.

Where can we find a molar mass? The mass spectrum - the peak with the highest m/z value corresponds to the molecular ion, and therefore the m/z value of that peak will equal the molar mass of the molecule[130], which is **88**.

Thus, the molecular formula of the molecule is $C_4H_8O_2$.

Question 6 (f) [Go to **Model Solution**]

What sort of information in the ^{1}H NMR spectrum will give you an idea as to the number of H atoms in each chemical environment? The **peak areas**!

For example, if peak A has an area of 0.45 units and peak B has an area of 1.35 units, it means that there are 3 times as many H atoms in the chemical environment corresponding to peak B as there are in that for peak A.

Applying this principle to the question, it is stated that there is **1** H atom in the chemical environment corresponding to the singlet at 11.9 ppm. The area of this peak is 0.07. Now, if we look at the doublet (which is what the question is referring to), the peak area is 0.41, pretty much **6** times the area of the first peak. Therefore, there must be six times the number of H atoms in the chemical environment corresponding to the doublet compared to that of the singlet (which has 1 H atom). Therefore, the answer is **6** H atoms.

Question 6 (g) [Go to **Model Solution**]

Question 6 (h) [Go to **Model Solution**]

In a scenario where you are given a formula for the compound as well as some IR and NMR spectra, and asked to deduce the structure, the first thing you should always do is **determine the number of double bond equivalents in the molecule**.

How many double bonds are present in this compound? Using the double bond equivalent formula, we find that there is **one** double bond or ring in this compound:

$$\begin{aligned} \text{DBE} &= \frac{2(\text{n(C)}+1)-\text{n(H)}+\text{n(N)}}{2} \\ &= \frac{2\times 5-8}{2} \\ &= 1 \end{aligned}$$

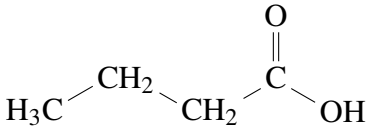
Looking at the IR spectra, there is a tell-tale sign of the presence of a C=O group - a strong absorption band at ~ 1700 cm^{-1}. This is one thing you should always look for in an IR spectrum - the presence of a strong absorption band at ~1700 cm^{-1}. Doing enough problems would make you familiar with the band, so you would not need to refer to the data book!

Since we have established that there is a C=O group (contains a double bond) there cannot be any more double bonds in this compound.

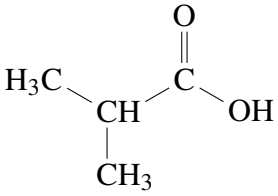
Looking at the NMR spectrum, we see a singlet at 12 ppm. From this alone we can say that there is a carboxy group, and this is supported by the presence of the C=O we found earlier!

[130]Note that there is a **tiny** peak at m/z = 89, which is likely attributed to a single ^{13}C isotope (1% of all C atoms are the ^{13}C isotope).

Now, we have to arrange the other three carbon atoms bonded to the carboxy group. Since there are no more double bonds, we know that the compound is either of the following:



Compound A

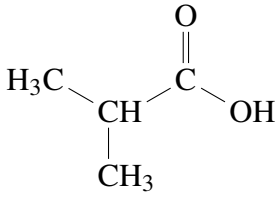


Compound B

There are two possible approaches you can use here:

- you can identify that there has to exist **6** H atoms in a single chemical environment, which is consistent only with Compound B above
- you can identify that there is a doublet present in the NMR spectrum indicating the presence of a carbon atom bonded to only one hydrogen atom (i.e. a CH group) - by the $n+1$ rule[131] - and only Compound B has a CH group

Hence, the answer is below:



To name the molecule - go systematically:

- there are **3** C atoms in the main carbon chain[132]; the affix here is "**prop**"
- there are no C=C double bonds; the affix used here is **an**
- there is a carboxyl (COOH) principal functional group; the affix is **-oic acid**
- there is a methyl (CH_3) group on carbon-2; the affix here is **methyl** (as a methyl group **must** be on carbon-2, so saying **2-methyl** is unnecessary)

The name of the molecule is therefore **methylpropanoic acid**.

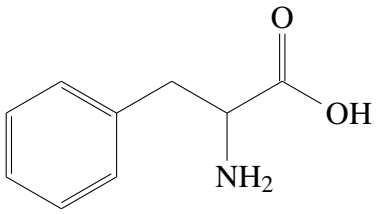
[131] Remember the n + 1 rule: If the number of other hydrogen atoms within three bonds of a particular hydrogen atom is n, the peak for that particular hydrogen atom would be split into $n+1$ peaks. Exceptions include hydrogen atoms bonded to nitrogen or oxygen.

[132] Don't forget the COOH carbon atom!

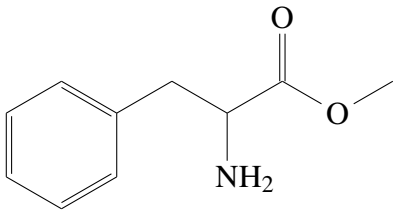
Question 7 (a) [Go to **Model Solution**]

Question 7 (b) [Go to **Model Solution**]

To figure out what **A** and **B** are, we should compare the structure of the main reactant (phenylalanine) and product (phenylalanine methyl ester):



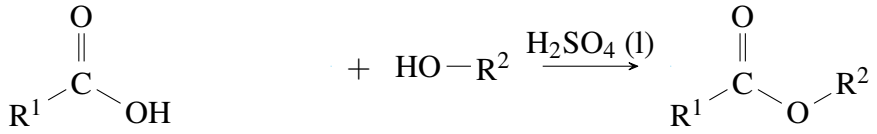
phenylalanine



phenylalanine methyl ester

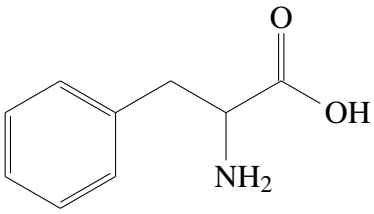
As you can see, the difference is at the right-hand side of the molecule. In phenylalanine, you have a **carboxyl** (COOH) group. In phenylalanine methyl ester, you have a (methyl) **ester** ($COOCH_3$) group. How do you convert a **carboxyl** group into an **ester** group?

Remember!
When a **carboxyl** group reacts with a **hydroxyl** group in the presence of concentrated sulfuric acid as a catalyst, you form an **ester**:

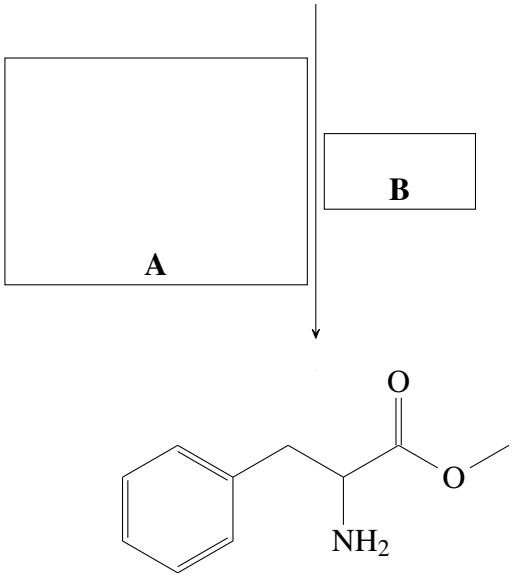


Therefore, reagent **A** must have a hydroxyl group. The question is - exactly what compound would compound **A** be? You can see that phenylalanine methyl ester (right) has an extra carbon atom next to the ester group; this extra carbon presumably comes from reagent **A**. Therefore, we can sy that reagent **A** has one carbon atom. What sort of molecule has only one carbon atom and a hydroxyl group? That molecule is **methanol** (CH_3OH)!

Hence, compound **A** is **methanol** and reagent **B** would presumably be the catalyst **concentrated sulfuric acid**.

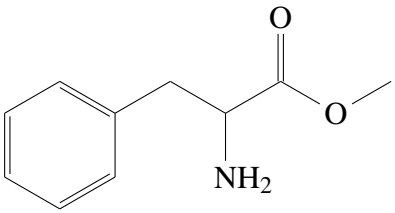


phenylalanine

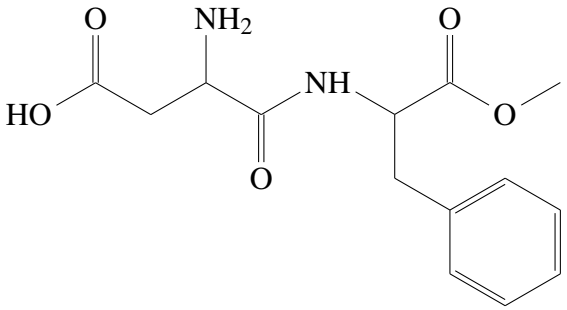


phenylalanine methyl ester

Now, the question is - what is compound **C**? We know that phenylalanine methyl ester reacts with compound **C** to form aspartame. Let us compare the structures of **phenylalanine methyl ester** and **aspartame**:

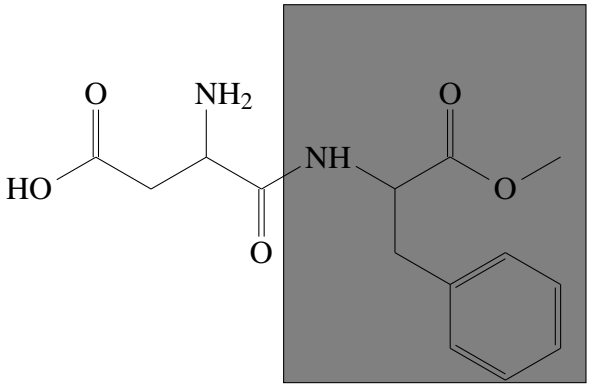


phenylalanine methyl ester



aspartame

Aspartame looks like a much bigger molecule. In fact, it seems to have a part that looks like phenylalanine methyl ester, as shown below. You'll find that apart from the NH_2 being replaced with NH, the shaded part is identical to phenylalanine methyl ester, just at a slightly different spatial arrangement:

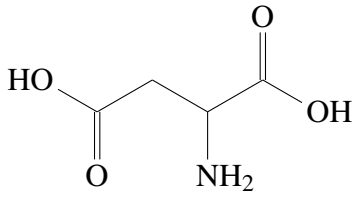


This means that compound C should look pretty much identical to the unshaded region of the aspartame molecule. The question is - how can compound **C** and phenylalanine methyl ester link together to form aspartame?

Have a look at the linkage between the shaded and unshaded area, the CONH linkage. You should recognise this as an **amide** linkage, formed by the joining of a **carboxyl** (COOH) group and an **amino** (NH_2) group.

The amino group is presumably derived from the phenylalanine methyl ester. Therefore, compound **C** must contribute the carboxyl group.

Thus, compound **C** has the following structure:



Question 7 (c) [Go to **Model Solution**]

This question is recall.

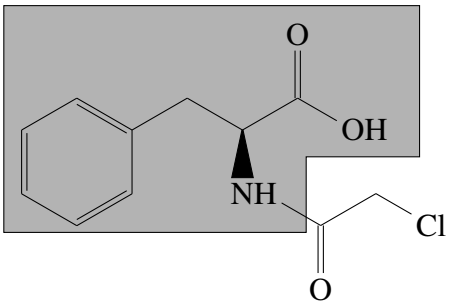
Question 7 (d) [Go to **Model Solution**]

Question 7 (e) [Go to **Model Solution**]

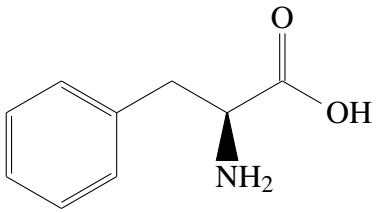
This solution covers this question part and the preceding question part.

The question states that the reaction is a **hydrolysis** reaction. From this alone, you should recognise that **water (H_2O)** is a reactant[133]. Compound **F** must be H_2O.

Let's compare the reactant L-chloroacetylphenylalanine and one of the products, L-phenylalanine.



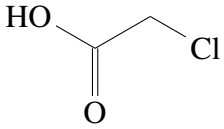
L-chloroacetylphenylalanine



L-phenylalanine

To find a location where hydrolysis can occur, look for a linkage of some sort in the original molecule, whether it be an ester, amide or glycosidic linkage. As you can see, there is an **amide** (CONH) linkage in the molecule, which can be hydrolysed to an **amino** group and **carboxyl** group. The amino group is part of the L-phenylalanine molecule as you can see above; the shaded area corresponds to the L-phenylalanine part of the original molecule. The carboxyl group must be part of the other compound **E**, whost structure should resemble the unshaded area of the original molecule.

Thus, the structure of compound **E** must be as follows:



Question 7 (f) [Go to **Model Solution**]

The clue here is that one of the reagents is porcine kidney acylase, which immediately tells us we are dealing with an enzyme[134].

Enzymes are protein molecules that catalyse biochemical reactions. When thinking about the words "enzyme" and "temperature" simultaneously, recall that enzyme function is temperature-dependent. Temperatures that are too high cause hydrogen bonds to break within the tertiary structure (and secondary), which change the conformation/shape of the enzyme molecule and therefore that of the active site. This means that the porcine kidney acylase enzyme is unable to bind to its substrate.

[133]The affix "hydro" means "water" and "lysis" means "to break". A hydrolysis reation is one in which water is used as a reactant to break a large molecule into smaller molecules.

[134]The dead giveaway is that the affix -ase is used to describe enzymes.

Thus, we can conclude that the temperature should not be above 37 degrees Celsius because this could render the porcine kidney acylase non-functional.

Question 7 (g) [Go to **Model Solution**]

Another property of enzymes you should recall is that they are **stereoselective**. This means that an enzyme molecule is able to bind significantly better to a specific enantiomer of a substrate as opposed to the other enantiomer, and thereby catalyse the reaction of primarily one of the enantiomers. This is because enzyme molecules themselves are composed of amino acids of only one enantiomer, and therefore the active site of the enzyme molecule has a shape that can only accommodate one enantiomer of the substrate. A useful analogy is that it is very difficult if not impossible to put your right hand into a left-handed glove.

In this case, to produce D-phenylalanine, you will need D-chloroacetylphenylalanine to bind to the porcine kidney acylase substrate. As it turns out, the enzyme probably binds L-chloroacetylphenylalanine to a significantly greater extent than it does D-chloroacetylphenylalanine, which would explain why only the L-enantiomer was produced upon enzyme catalysis.

Question 8 (a) [Go to **Model Solution**]

The lithium ion battery is a **secondary cell**. It acts as a galvanic cell during discharge, and as an electrolytic cell during recharge. Since we are talking about **recharge** in this question, consider this cell as an electrolytic cell at this moment.

Remember!
The polarities of the electrodes in a galvanic and electrolytic cell are shown below:

	anode (oxidation)	cathode (reduction)
galvanic cell	$-$	$+$
electrolytic cell	$+$	$-$

So, the positive electrode is the **anode**, and as per the AN OIL RIG CAT mnemonic, **oxidation** occurs at the anode.

Consider the two reaction equations that occur during **discharge**, as written in the question stem:

$$\begin{aligned} \mathrm{LiC_6} &\rightarrow \mathrm{Li^+ + e^- + C_6} \\ \mathrm{CoO_2 + Li^+ + e^-} &\rightarrow \mathrm{LiCoO_2} \end{aligned}$$

During recharge, the two reactions are **reversed** as electricity is shoved 'the wrong way' through the cell:

$$\begin{aligned} \mathrm{LiC_6} &\leftarrow \mathrm{Li^+ + e^- + C_6} \\ \mathrm{CoO_2 + Li^+ + e^-} &\leftarrow \mathrm{LiCoO_2} \end{aligned}$$

or, as more conventionally expressed:

$$\begin{aligned} \mathrm{Li^+ + e^- + C_6} &\rightarrow \mathrm{LiC_6} \\ \mathrm{LiCoO_2} &\rightarrow \mathrm{CoO_2 + Li^+ + e^-} \end{aligned}$$

Since the positive electrode is the anode, and oxidation occurs at the anode (in an electrolyic cell), it follows that since oxidation involves the loss of electrons, electrons must be released as a separate product in the half equation and therefore should be found on the RHS of the equation. Hence, the correct half equation is:

$$LiCoO_2 \rightarrow CoO_2 + Li^+ + e^-$$

Question 8 (b) [Go to **Model Solution**]

An aqueous electrolyte means that water is present in the electrolyte. Now, what would be the issue with **lithium metal** and **water** together? Lithium is a group 1 metal and it should instantly hit you that as a group 1 metal, lithium can react vigorously with water (even violently at times). The reaction is very exothermic and hydrogen gas is produced. This hydrogen gas can build up in the battery and cause it to explode, or if there were a defect in the battery, it can get into contact with air and combust in the high temperatures of the battery[135].

To write the reaction equation, you can find the two half equations corresponding to the oxidation of Li to Li^+ and the reduction of H_2O to H_2 on the electrochemical series:

$$Li\ (s) \rightarrow Li^+\ (aq) + e^- \quad (1)$$
$$2\ H_2O\ (l) + 2\ e^- \rightarrow H_2\ (g) + 2\ OH^-\ (aq) \quad (2)$$

To combine the redox half equations, multiply equation (1) by 2 so that both equations have 2 electrons that can cancel out:

$$2Li\ (s) \rightarrow 2\ Li^+\ (aq) + 2\ e^- \quad (3)$$
$$2\ H_2O\ (l) + 2\ e^- \rightarrow H_2\ (g) + 2\ OH^-\ (aq) \quad (2)$$

Combining the two equations:

$$2\ Li\ (s) + 2\ H_2O\ (l) \rightarrow 2\ Li^+\ (aq) + 2\ OH^-\ (aq) + H_2\ (g)$$

The following is also equivalent:

$$2\ Li\ (s) + 2\ H_2O\ (l) \rightarrow 2\ LiOH\ (aq) + H_2\ (g)$$

Question 8 (c) [Go to **Model Solution**]

This is a classic question asking how increasing the temperature increases the reaction rate.

Remember that increasing the temperature will cause the reactant molecules to, in general, **move faster**. Now, the reactant molecules moving faster and having higher kinetic energies means that when they collide, they will collide **'harder'** and with **higher energies**. Now, remember that only collisions with an energy that is equal to or higher than the **activation energy** will be successful. Increasing the temperature will increase the fraction of collisions that are equal to or above the activation energy. Hence, increasing the temperature will increase the fraction of collisions between reactant molecules that are successful.

In addition to this, because the reactant molecules are moving **faster**, they will **collide more frequently**.

[135] When you have spare time, go onto YouTube and look up "reaction of alkali metals with water" and you'll see quite a spectacle.

Therefore, increasing the temperature increases the **frequency of collisions** between reactants **AND** the **fraction of collisions between reactant molecules that will be successful -** hence increasing the frequency of successful collisions and thus the reaction rate.

Whenever answering questions regarding factors that affect reaction rate, try and make reference to the frequency of **successful collisions** where appropriate.

Question 8 (d) [Go to **Model Solution**]

We have a case where there is a direct connection between the two electrodes by means of a finger-like projection of the LiC_6 electrode made out of lithium metal. Now, if the two electrodes are in direct contact with one another, what happens? The electrons would travel directly from one electrode to the other (via the Li finger-like projection), instead of through the external circuit. For those studying physics, it's pretty much a short circuit.

When this happens, the chemical energy is no longer converted to electrical energy; it is largely converted to **thermal energy**.

Remember that the idea behind a galvanic cell is that the reactants cannot be in direct contact with each other, forcing electrons to travel between the two reactants via the external circuit; in this case the energy transformation is from **chemical** to **electrical**. If they do come into direct contact with one another, the energy transformation becomes **chemical to thermal**.

How would this case **thermal runaway**? The question stem states that thermal runaway is where the temperature increases at an ever increasing rate. Why does this happen? One clue is that there is significant production of heat from the conversion of chemical energy to thermal energy. But this does not explain why the temperature increases **at an ever increasing rate**. What is the other half of the story?

One good trick in exams is that sometimes previous question parts give a clue as to the answer in future question parts. Part **c** asked you to explain why an increased temperature...increases the reaction rate. Well. This could be the other half of the story. Suppose the electrodes get into contact, heat is released and...the temperature increases. The implication of that is that the rate of the reaction between the substances in the electrodes would increase...releasing heat more rapidly, increasing the temperature. This is a vicious cycle which...characterises thermal runaway.

Question 8 (e) [Go to **Model Solution**]

This question is quite subtle. What could cause an explosion in a battery generally? It could be caused by the rapid buildup of a gas. It could be caused by a violent chemical reaction. We need to get some clues from the question stem. Let's read the question and question stem carefully.

Explain why using too high a current to recharge such a phone can lead to an explosion.

What have we established in previous question parts about using too high a current to recharge? We know that this causes metallic lithium to be plated on the surface of the negative electrode, which leads to dendritic projections that contact the other electrode, leading to this release of heat and a vicious cycle (as previously discussed) making the battery extremely hot. Thus, we know that "high current on recharge" means "**very hot battery**".

Can we figure out any consequences? Well, the question stem states:

> It is known that at very high temperatures some of the components of the battery can decompose...firstly, the organic solvent can decompose into small hydrocarbon molecules (eg. **ethane**).

Ethane being released in the battery? That's kind of scary. Ethane is a gas, and it is a known fuel. Any contact with oxygen and the battery is royally screwed. Arguably, the buildup of ethane itself can lead to an explosion from the high pressures generated. Let's read on.

> ...secondly, the cobalt oxide "part" of the lithium cobalt oxide electrolyte can decompose as per the following equation...
>
> $$3\ CoO_2 \rightarrow Co_3O_4 + O_2$$

Any contact with oxygen. At this stage, certain four-letter words in your vocabulary could be of some use here[136]. Ethane and oxygen mixed in the battery...at high temperatures...calls for a giant explosion as ethane combusts.

Question 9 (a) [Go to **Model Solution**]

To review the concept of viscosity, click **here**.

> **Remember!**
> If the intermolecular bonding within the fluid is strong, the fluid will tend to be viscous.

Therefore, in this question, we need to compare the degree of intermolecular bonding within methanol and butan-1-ol.

Let us draw their structures:



methanol

butan-1-ol

We need to consider the following types of intermolecular bonding that are present in each substance and determine which substance would exhibit stronger intermolecular bonding overall.

> **Remember!**
> The types of intermolecular bonding (bonding/attraction **between** molecules) that exist are:
>
> - dispersion forces
> - dipole-dipole interactions
> - hydrogen bonds

[136] Preferably not out loud. Especially not during the exam.

This gives rise to the following table:

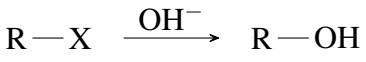
Intermolecular bond	**Methanol**	**Butan-1-ol**
hydrogen bonding	Present due to the -OH groups	Present due to the -OH groups
dipole-dipole interactions	Not present (aside from the hydrogen bond already described).	Not present (aside from the hydrogen bond already described).
dispersion forces	Weaker, as methanol is a smaller molecule.	Stronger, as butan-1-ol is a larger molecule.

As you can see, whilst both molecules have similar strength hydrogen bonds, butan-1-ol will have stronger dispersion forces. Therefore, as a whole, butan-1-ol will have the stronger intermolecular bonding and, therefore, be more viscous.

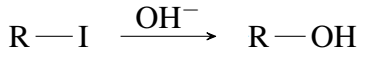
Question 9 (b) [Go to **Model Solution**]

The structural formula of the iodoalkane already has an inherent clue in it. It is an **iodo**alkane. Presumably there will be at least one iodine (I) atom in the molecule as a substituent.

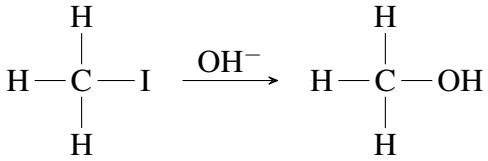
So, this reaction will convert an iodoalkane, which is a haloalkane, to an alcohol. This should remind you of the following substitution reaction:



This would apply to an iodoalkane:

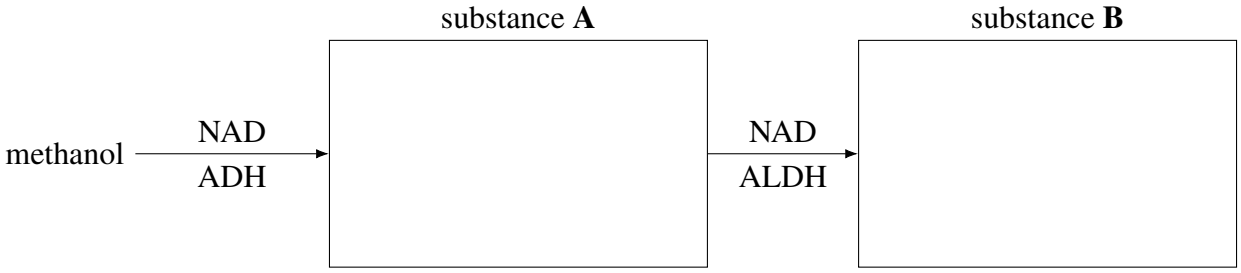


Therefore, we can come up with the following:



Question 9 (c) [Go to **Model Solution**]

The idea here is to identify substance A and B. Let us review the diagram given in the question stem:



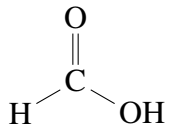
Let's review the preceding text to draw clues about the identities of substances A and B:

> In humans, there are two enzymes, namely ADH and ALDH, that catalyse the oxidation reaction of methanol and its intermediate products. Also, human cells contain significant quantities of **NAD (nicotinamide adenine dinucleotide) which is an oxidising agent**[a] that acts as an additional substrate for ADH and ALDH.
> **Methanol is initially oxidised in the presence of ADH to produce substance A**[b], which itself is **subsequently oxidised in the presence of ALDH to produce substance B**[c]. It is known that substances **A** and **B** are both liquids at room temperature and if substance **B** were dissolved in a **solution of sodium bicarbonate ($NaHCO_3$), bubbling**[d] would be observable.
>
> [a] So, NAD is an oxidising agent.
> [b] Methanol is **oxidised** to form substance A. Think about what you have learned about how alcohols can be oxidised.
> [c] A is oxidised to form B.
> [d] Mixing this chemical with sodium bicarbonate (i.e. baking soda) would lead to the formation of bubbles. Isn't this what acids do when they react with hydrogencarbonate ions? The gas produced is CO_2.

With this in mind, we can say that:

- **Substance B.**

  Substance B must be an acid that is similar in structure to methanol; it must be a 1-carbon compound. There is only one organic acid that is a one-carbon compound, which we can draw below:

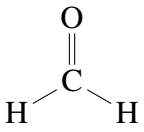
  

  methanoic acid

  We know that methanol can be oxidised to form methanoic acid, as **primary alcohols can be oxidised to form carboxylic acids**.

- **Substance A.**

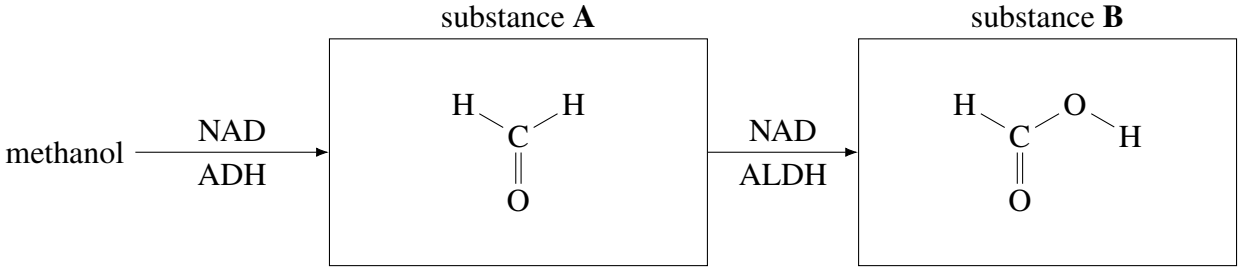
  Don't forget that primary alcohols are oxidised to carboxylic acids **via an intermediate product**; they are oxidised to this intermediate product first, which is then oxidsed to form the carboxylic acid. This intermediate product is an **aldehyde**. On this occasion, this would be a 1-carbon aldehyde, also known as methanal:



methanal

This must be structure A.

Remember that the question stem asked you to show all bonds. This should give you the following:



Question 9 (d) [Go to **Model Solution**]

This question is basically just asking you why enzyme function is best at a specific temperature; this is practically recall.

Remember, the ability of an enzyme to function is dependent on its ability to bind to its substrate, which itself is dependent on the shape of the active site of the enzyme being complementary to that of the substrate (like a lock and key). Let's look at what would happen to the enzyme at low, moderate and high temperatures.

Low temperature.

Here, the molecules are moving very....slowly....like a sloth[137]. That is, the enzyme and substrate molecules.

This means that the frequency of collisions between the enzyme and substrate molecules would be rather low and, of those, only a fraction of those would be in the right orientation such that the substrate smashes straight into the active site of the enzyme molecule. In addition to this, only a fraction of the collisions between the enzyme and substrate molecules would have sufficient energy behind them[138] to break through the (low) activation energy barrier[139].

In essence, the rate of formation of enzyme-substrate complexes and, hence, the rate of product formation, will be slow. In other words, the enzyme activity is rather low here.

Moderate temperature.

As the temperature increases, the enzyme and substrate molecules move faster. They collide more often (and a fraction would collide in the right orientation), and on average, they collide "harder" with one another. The rate of formation of enzyme-substrate complexes increases and, therefore, the rate of product formation increases. With an increase in temperature, the enzyme activity increases.

[137] Or me when I try to get out of bed to go to work. Yes, that feeling doesn't go away once you finish school. If you're anything like me that is. Just ask my wife.

[138] In simple terms, the enzyme and substrate molecules must be moving with sufficient speed so that they can slam into one another "hard enough".

[139] This sounds like a reaction rates concept. That's because it is a reaction rates concept.

High temperature.

You could argue that you could just keep increasing the temperature, and the enzyme activity would just increase. This is true, until you reach a critical temperature - and this critical temperature depends on the enzyme. Once you get past this critical temperature, the enzyme molecules are moving so fast and are vibrating so hard that they start **changing their shape** as **hydrogen bonds** within their secondary and tertiary structures break; the hydrogen bonds, whilst strong, are not strong enough to withstand the intense vibration of the hot enzyme molecule.

Remember that it is the secondary and tertiary structures of the enzyme molecule that determine their 3D shape. If these are being altered by the breakage of hydrogen bonds, the 3D shape of the enzyme changes and, with that, the 3D shape of its active site.

The active site becomes increasingly unable to bind properly to the substrate and, therefore, the enzyme loses its ability to act as an enzyme. The enzyme is **denatured**.

Just apply this theory to the question.

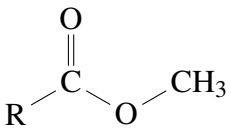
Question 9 (e) [Go to **Model Solution**]

This question is essentially recall. Remember that biodiesel is sourced from vegetable oil.

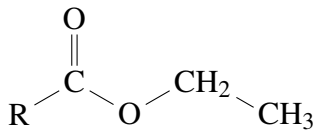
Biofuels are fuels that are produced from living or recently deceased plant or animal matter.

Question 9 (f) [Go to **Model Solution**]

It may be worth reproducing some of the Detailed Solution from earlier in the book. **Biodiesel** is a biofuel that is composed of modified fatty acids - namely simple esters of fatty acids. The common ones you'll encounter will have a methyl (CH_3-) or ethyl (CH_3CH_2-) group at the end, like so:



fatty acid methyl ester



fatty acid ethyl ester

Here, the R- group is a fatty acid tail, essentially the bit of the fatty acid that is not the COOH bit.

In short, biodiesel molecules are **esters**. Remember that esters can be synthesised by reacting **carboxylic acids** (eg. linolenic acid) with **alcohols** (eg. methanol) in concentrated sulfuric or phosphoric acid. This should give you the following equation:

$$C_{17}H_{29}COOH\ (l) + CH_3OH\ (l) \xrightarrow{H_2SO_4} C_{17}H_{29}COOCH_3\ (l) + H_2O\ (l)$$

Question 9 (g) [Go to **Model Solution**]

Let us look at the aim first:

> to determine whether the methanol is contaminated with small quantities of ethanol and to **quantify** the degree of contamination

So, the aim is to determine **how** contaminated the methanol is. Just knowing that it is contaminated is not good enough. We want to know how contaminated it is. This is an example of quantitative analysis.

Let's look at each technique.

High performance liquid chromatography.

Here, a mixture of chemicals are added to the top of a glass cylinder (column) filled with - essentially - sand. Very fine sand. A liquid solvent (known as the eluent) is then forced through the column at exceptionally high pressure, carrying the mixture of chemicals with it.

This separates out the chemicals according to their relative ability to dissolve in the eluent versus their relative ability to **adsorb**[140] onto the sand grains (also known as the stationary phase). As the chemicals exit the column, separated out, they pass through a detector which detects the amount of each chemical passing through the column and produces a signal. The area under the curve generated by this signal (when plotted against time) is proportional to the amount of each chemical passing through the detector. We can use this area, compare it against areas generated by known amounts of the chemicals of interest generating a calibration curve, to determine the amount of each chemical in the mixture.

In this scenario, it would be possible to separate out methanol and ethanol, being molecules of different size and therefore having different dispersion force strength - the ethanol would have a slightly greater ability to adsorb onto the stationary phase. It would also be possible to quantify the amount of ethanol that is contaminating the methanol sample by measuring the area under the curve corresponding to methanol.

Mass spectrometry.

This is good at determining the size of molecules, since you can look at the peak with the largest m/z value to determine the molar mass of the molecule. However, mass spectrometry is not particularly good at determining **how much** of a particular chemical there is in the sample. It could tell, though, if the methanol has been contaminated, since the presence of a peak with m/z value of 46 would be a dead giveaway as to the presence of ethanol (46 is the relative molecular mass of ethanol whereas that of methanol is 32, meaning pure methanol cannot give you a mass spectrum containing a peak with m/z value of 46).

Given the aim is to quantify how much ethanol is in the sample of methanol, we would easily pick HPLC.

Question 10 (a) [Go to **Model Solution**]

The first thing to do in this setting is to figure out which reactant is being reduced and which is being oxidised. Here, permanganate (MnO_4^-) is reacting with ethanol (C_2H_5OH) to form manganese (II) ions and ethanoic acid. Which is being reduced, MnO_4^- or C_2H_5OH? The best way to figure out if a species is being oxidised or reduced is to compare the **oxidation number** of the key atom in the reactant and the corresponding product.

For MnO_4^-, the corresponding product would be Mn^{2+} since they are the only reactant and product respectively that contain Mn atoms. Let's figure out the oxidation number of Mn in both of these:

[140] Adsorb means "to stick onto".

- **MnO_4^-**

  Oxygen, being an extremely electronegative atom and forming two bonds generally, will most definitely have an oxidation number of -2.

  Since the sum of the oxidation numbers must be -1, it follows that, supposing the oxidation number of Mn is x:

  $$x + 4 \times (-2) = -1 \Rightarrow x - 8 = -1 \Rightarrow x = 7$$

  Hence, the oxidation number of Mn is +7.

- **Mn^{2+}**

  It is self explanatory that the oxidation number of Mn here is +2.

Therefore, the oxidation number of Mn has dropped. Hence, MnO_4^- is being reduced and therefore the reduction half equation would involve MnO_4^- and Mn^{2+}. Now, do we do this in acidic or basic solution? The best indication is from the question stem.

Looking at the question stem, the reaction equation for the overall reaction features H^+ as a reactant. H^+ ions can only exist in significant concentrations in acidic solutions. Therefore, we need to write down the reaction half equation for the reduction of MnO_4^- in **acidic** solution. To do this, we use the **KOHES** method.

Balancing half-equations in acid
We assume that there are ample amounts of H^+ in solution, given this is an acidic solution.
The KOHES method is below:

1. Balance **K**ey atoms.
2. Balance **O** atoms by adding H_2O molecules to the appropriate side of the reaction equation.
3. Balance **H** atoms by adding H^+ ions.
4. Balance charge by adding **E**lectrons to the appropriate side of the reaction equation.
5. Add **S**tates.

- Balance **K**ey atoms.

  $$MnO_4^- \rightarrow Mn^{2+}$$

  The Mn atoms are already balanced.

- Balance **O** atoms by adding H_2O molecules to the appropriate side of the reaction equation.

  To balance O atoms, we add H_2O to the appropriate side since the reaction is in solution and therefore there will definitely be H_2O molecules swimming around. There are 4 O atoms on the LHS, and therefore to balance this with 4 O atoms on the RHS, we add 4 H_2O molecules to the RHS.

  $$MnO_4^- \rightarrow Mn^{2+} + 4\ H_2O$$

- Balance **H** atoms by adding H^+ ions.

  To balance H atoms, we add $H^+$to the appropriate side on this occasion, because if the reaction occurred in acidic solution, there'd be a significant amount of H^+ ions swimming around in solution. There are 8

H atoms on the RHS thanks to the H_2O molecules we added on the RHS. To balance this out, we add 8 H^+ ions to the LHS.

$$MnO_4^- + 8\ H^+ \rightarrow Mn^{2+} + 4\ H_2O$$

- Balance charge by adding **E**lectrons to the appropriate side of the reaction equation.

  We need to balance **charge** here by adding electrons to the appropriate side.

  The total charge on the LHS (before adding electrons) is $-1 + 8 = +7$, whereas that on the RHS is +2. Therefore, we need to add electrons to the **LHS** to **decrease** the charge on the LHS (remember electrons are **negatively** charged) to make it equal to that of the RHS (+2). Therefore, we add 5 electrons to bring the LHS charge from +7 to +2.

$$MnO_4^- + 8\ H^+ + 5\ e^- \rightarrow Mn^{2+} + 4\ H_2O$$

- Add **S**tates.

$$MnO_4^-\ (aq) + 8\ H^+\ (aq) + 5\ e^- \rightarrow Mn^{2+}\ (aq) + 4\ H_2O\ (l)$$

Question 10 (b) [Go to **Model Solution**]

We need to work out the concentration, in mol L^{-1}, of ethanol in the vodka.

We have a **50.00 mL** sample of vodka (a volume). Therefore, we need to work out the amount, in mol, of ethanol in this 50 mL sample of vodka to complete the question.

Looking at the question stem, we had **50.00 mL** of vodka, which was diluted to **500.0 mL** with water. Of this 500.0 mL, **20.00 mL** aliquots were taken for the titration. Therefore, the amount of ethanol in each of these aliquots would be a fraction (20/500ths) of the total amount of ethanol in the original vodka. The only indication we have as to the amount of ethanol in the vodka is the titration with the permanganate, where the 20.00 mL aliquots of diluted vodka were used. Hence, we should determine the amount of ethanol in the 20.00 mL aliquot.

We do know the concentration (0.5022 M) and the average volume (23.22 mL - the average titre) of permanganate solution to react with all the ethanol in the 20.00 mL diluted vodka. Therefore, we can work out the amount, in mol, of permanganate ions in the titre:

$$n(MnO_4^-) = 0.5022 \times 0.02322 = 0.01166 \text{ mol}$$

Looking at the reaction equation, we see that permanganate and ethanol react in a 4:5 mole ratio. That means, **4 moles** of permanganate would be required to completely react with **5 moles** of ethanol. However, we only needed 0.01166 mol of permanaganate to react with all the ethanol in the 20.00 mL of diluted vodka. Thus, the amount of ethanol in the diluted vodka is

$$n(C_2H_5OH)_{\text{aliquot}} = 0.01166 \times \frac{5}{4} = 0.01458 \text{ mol}$$

Now that we know the amount of ethanol in the 20.00 mL aliquot of diluted vodka, what about the total 500.0 mL of diluted vodka? Well, if 20.00 mL of diluted vodka contained 0.01458 mol of ethanol, the amount of ethanol in 500.0 mL of diluted vodka is

$$n(C_2H_5OH)_{\text{total}} = 0.01458 \times \frac{500.0}{20.00} = 0.3644 \text{ mol}$$

How much ethanol, therefore, was there in the original undiluted 50.00 mL sample of vodka? Remember that we derived the 500.0 mL solution by just adding water to the 50.00 mL of vodka to top it up to 500 mL - so we did not add or remove any ethanol from the vodka. Thus, the amount of ethanol in the undiluted 50.00 mL vodka is the same as that in the diluted sample, which is 0.3644 mol.

Now that we have the amount of ethanol in the 50.00 mL sample of vodka, let us work out the concentration of ethanol:

$$[C_2H_5OH] = \frac{0.3644}{0.05000} = 7.288 \text{ M}$$

Question 10 (c) [Go to **Model Solution**]

The first thing you should recall is that %(v/v) means the number of **mL** of solute in **100 mL** of solution. Therefore, the question is asking - how many mL of ethanol is there in 100 mL of the vodka?

The second thing to note is that the density of ethanol is 0.789 g mL^{-1}. This means that 1 mL of ethanol weighs 0.789 g. Hence, if we know the mass of pure ethanol in some sample, we will know its volume as well, and vice versa.

We know that there is **7.288 mol** of ethanol in **1 L (1000 mL)** of vodka. We need to work out the volume, in mL, of ethanol in only **100 mL** of vodka. Now, if we know the amount, in mol, of ethanol in 100 mL og vodka, we could quite easily find its mass. Since we know the density of ethanol, we can easily convert mass of ethanol to volume of ethanol, which is what we need to complete the question.

So, since there is 7.288 mol of ethanol in 1 L of vodka, we can say that there are 7.288×46.0 = 335 g of ethanol in 1 L of vodka. Note that 46.0 g mol^{-1} is the molar mass for ethanol.

Now that we know the mass of ethanol in 1 L (1000 mL) of vodka, we can easily find out the mass of ethanol in ony 100 mL of vodka, which would be just 33.5 g.

So now that we know the mass of ethanol in 100 mL of vodka, the only thing we need to do to get the answer is to convert the mass of ethanol into volume of ethanol, since we need to work out the volume of ethanol in 100 mL of vodka.

Since 0.789 g of ethanol would occupy 1 mL, what volume would 33.5 g occupy?

$$\begin{aligned} c(\text{ethanol}) &= 33.5 \text{ g } (100 \text{ mL})^{-1} \\ &= \frac{33.5}{0.789} \text{ mL } (100 \text{ mL})^{-1} \\ &= 42.5 \text{ mL ethanol } (100 \text{ mL vodka})^{-1} \\ &= 42.5\% \text{ (v/v)} \end{aligned}$$

This can sometimes be a tricky calculation. Here is a quick way of logicking it out:

- I have 33.5 g of ethanol and want to work out its volume.
- I know the density of ethanol is 0.789 g mL^{-1} (1 mL of ethanol weighs 0.789 g).
- Either of the following ways of reasoning it out are fine:

- I either have to multiply by 0.789 or divide by 0.789 (33.5×0.789 or $33.5 \div 0.789$). Now, if 1 mL of ethanol weighs 0.789 g, we would expect the volume of ethanol to be greater than its mass (the numerical value). So hence, since the mass of ethanol is 33.5 g, we would expect the volume to be more than 33.5 mL. If we multiply 33.5 by 0.789 (a number less than 1), we would get a number less than 33.5. If we divide 33.5 by 0.789, it would give us a number bigger than 33.5. Therefore, we probably have to divide by 0.789.
- You can **play with units**. If we divide **g** by (**g mL**$^{-1}$), we should get mL. To demonstrate:

$$\begin{aligned} \frac{\text{g}}{\text{g mL}^{-1}} &= \frac{1}{\text{mL}^{-1}} \\ &= \text{mL} \end{aligned}$$

Therefore, you can see the calculation this way:

$$\begin{aligned} \text{answer} &= \frac{33.5 \text{ g}}{0.789 \text{ g mL}^{-1}} \\ &= \frac{33.5}{0.789} \frac{\text{g}}{\text{g mL}^{-1}} \\ &= 42.5 \text{ mL} \end{aligned}$$

Question 10 (d) [Go to **Model Solution**]

This extended response question draws upon your knowledge and experience in doing your practical investigation scientific report. The skills you learnt in this SAC will be needed to solve this question. This question is basically a scenario where you are writing a scientific report.

To be able to do this question, it would be helpful to plan your response and jot down some dot points. So, let's have a look at what specific pieces of information the examiner is looking for.

- summarise the **results** that were obtained
- assess the **reliability** and **validity** of the results obtained[141]
- suggest **reasons** as to why the data may or may not be accurate, and explanations for **interesting observations**
- suggest **improvements** to the experimental design

Let us go through each of these in turn.

Summarise the results that were obtained

From part **c**, you have calculated that the ethanol content in the Grey Goose Vodka was 42.5% (v/v). Now, how is this significant? Why did we do this experiment in the first place? Why do we care? One reason is because the manufacturer said that the ethanol content was 40.0% (v/v). Perhaps the manufacturer was BSing (don't write that in your report though). Anyway, when summarising the results that you obtained, it is helpful

[141] I do have to apologise here. The term "reliability" has been removed from the Study Design because it is an ambiguous term. A better word to use in place of "reliability" is "precision". Also, it would be better to refer to "accuracy" rather than validity". You can be lenient with yourself a little here if you are marking yourself.

to make a comparison between your hypothesis and your actual result. Therefore, you would say something along the lines of:

> The ethanol content of the vodka was determined to be 42.5% (v/v), higher than the manufacturer's quoted value of 40% (v/v).

Now, onto the next section.

Assess the reliability/precision and validity/accuracy of the results obtained.

To address this particular dot point, you need to know what the terms **reliability/precise** and **accuracy** mean. Below is a discussion on these two terms:

- Data that is **reliable/precise** is data that is **consistent** with one another. In other words, when you do multiple trials or repeat experimental setups, if the results you get for the trials are very close to one another, then the data is reliable. For instance, if you have titre volumes (in repeat experimentals setups) of 23.22 mL, 23.22 mL, 23.23 mL, 23.21 mL, 23.22 mL - this data is precise. Data that is precise has minimal **random error**. However, note that precise data is not necessarily accurate.
- Data that is **valid/accurate** is data that (when averaged) gives you a value that is very close (if not the same) to the true value. For instance, if the real concentration of some HCl solution was 0.1000 M, and with multiple titrations you get values of 0.1004 M, 0.1006 M, 0.0996 M and 0.0994 M (which average to 0.1000 M), this data would be considered valid. Data that is valid has minimal **systematic error**. However, again note that valid data is not necessarily precise.
- Data that is **precise but not accurate** is data where the values you get have very **low variability** but **average to a value other than the true value**.
- Data that is **accurate but not precise** is data where the values have **high variability** but **average to (or close to) the true value**.

What about in this case?

- **Precision**

  We can see that, despite the outlier that we dismissed as a non-concordant titre, the titre volumes obtained were very consistent with one another. Therefore, this data is definitely precise.

- **Accuracy**

  Is this data accurate? We wouldn't know for sure. Yes, we got a value that is above the quoted manufacturer's value. That means either the manufacturer is BSing to us or has messed up, or we messed up. The question to ask here is - **are there any possible sources of systematic error in this experiment**? This is addressed in the next section.

Suggest reasons as to why the data may or may not be accurate, and explanations for interesting observations.

This section attracted the most marks in this question (a total of 8 marks out of 10). This is where the meat is. Now, finding reasons why the data may potentially be inaccurate means that we have to look for sources of **systematic error** - error that is intrinsic to the methodology of the experiment which will consistently occur every time you perform in the experiment. Note that systematic error is **NOT** human error; human error is

where the scientist makes a mistake when performing the experiment. Systematic error is the error you **consistently** get, if you perform the experiment as per the methodology correctly.

Finding sources of systematic error can be a bit of a pain. The **best way** to get good at this is to draw on your practical experience doing pracs at school. Alternatively, if you haven't had the chance to develop these skills, doing lots of experimental design problems will hone your skills in picking holes in experiments.

All types of experiments have common sources of systematic error, from redox titrations to calorimetery. For **redox titrations**, common sources of systematic error that you should remember and watch out for in the question are:

- Titrants are **non-specific**. In other words, when you add say an oxidising agent (as the titrant) into the conical flask during a titration, it could potentially react with many different reducing agents in the conical flask, not just the chemical you are analysing. This is particularly the case when you add a **powerful oxidising agent** as the titrant, such as permanganate. In this case, when you add permangante into the conical flask, it will certainly react with the ethanol (as a reducing agent). But the vodka is a mixture of substances. How do you know that there are not other chemicals in the vodka that are also reducing agents and could react with the powerful oxidising agent permanganate? If there were such other chemicals, then it would take a higher volume of permanganate solution to reach the endpoint and you would get an **overestimate** of the ethanol content in the vodka.
- At Year 12 level, it is easy to **overshoot the endpoint by at least a slight amount**. This is because when you **add the titrant dropwise** into the conical flask - each drop has a volume of about 0.05-0.10 mL. This means that you are adding titrant in increments of 0.05-0.10 mL. Therefore, when you add the last drop of titrant to get the endpoint, you may have crossed the equivalence point and overshot it a little in that single drop. To mitigate this, you can add half- or quarter drops using a technique that experienced chemists use.
- Sometimes in redox titrations, the endpoint will be **just after** the equivalence point. This issue is important in this case. This is because permanganate is a very purple solution and the products are clear. This means that the colour change, from clear to purple, will only occur when there is a small excess of permangante in the flask that does not react.

In summary, a basic screen of common issues[142] with redox titrations you can use are:

- titrants can react with other chemicals in the conical flask
- the endpoint can be overshot very easily due to using a dropwise technique
- the endpoint may be intrinsically just after the equivalence point

In this particular case, the basic screen has given us three issues:

- The permanganate can react with other chemicals besides ethanol in the conical flask, which will lead to a higher titre volume and an overestimate of the ethanol content in the vodka
- the use of a dropwise technique could lead to slight overshooting of the endpoint, again leading to a higher titre volume and overestimate of ethanol content.

[142] Note that these are just common issues; they may not all necessarily be present in all cases!

- The endpoint is slightly after the equivalence point, meaning the titre volume will be higher than it should be, with similar outcomes.

Another issue that is specific to the titration is given in the question stem, if you read carefully:

> In the presence of a powerful oxidising agent, ethanoic acid can be oxidised to carbon dioxide.

Now this is interesting. How is this significant? Is ethanoic acid even involved in this question? Well, remember that ethanol can be oxidised to ethanoic acid (from your organic chemistry you learnt this year). This should prompt you to look at the reaction equation:

$$4\ MnO_4^-\ (aq) + 5\ C_2H_5OH\ (aq) + 12\ H^+\ (aq) \rightarrow 4\ Mn^{2+}\ (aq) + 5\ \mathbf{CH_3COOH\ (aq)} + 11\ H_2O\ (l)$$

Well. So during the titration, you'll have ethanoic acid swimming around in the conical flask. What if some of the ethanoic acid reacted with extra permanganate to form CO_2? Is there any indication that this occurred during the titration? Let's have a look at the logbook notes for any clues:

> titration #1 - add first drop of titrant, contents of reaction flask turn pink immediately, then after few seconds contents turn clear. add second drop of titrant, contents turn pink and immediately turn clear. add titrant in a slow stream, contents remain a very slight pink colour, stop stream - contents turn clear immediately. ***bubble formation observed on surface of solution.*** *titrant then added dropwise until colour change persistent for 10 seconds.*

Well, that's a bit odd. Bubble formation in a reaction that is not expected to produce gas as a product as per the reaction equation. What if that gas was CO_2 from the oxidation of ethanoic acid by permanganate?

What could this mean? If this were the case, it means that you would need to add extra permanganate solution to reach the endpoint (because some of the permanagate solution would be diverted to react with ethanoic acid to form CO_2 in what is likely a slower reaction). This would mean a higher titre volume and therefore an overestimate of the ethanol content in the vodka!

Now, to get around this issue - the question said that:

> ...in the presence of a **powerful oxidising agent**, ethanoic acid can be oxidised...

Therefore, you could use a weaker oxidising agent than permanganate.

Those would be the main systematic errors I could think of.

Now, based on these potential systematic errors - is it possible that the data is inaccurate? Quite certainly. Therefore, we cannot guarantee the accuracy of the data. The data could be accurate; the slight overshooting of the endpoint would only lead to a very very small increase in the calculated ethanol content, there may not have been any other reducing agents in the conical flask and the amount of CO_2 produced may be extremely small - or the bubbles formed could be from the agitation of the solution during the titration rather than the formation of CO_2. However, the point is that we cannot guarantee the validity of the data.

Now, how about interesting observations? Well, we find observations in the logbook entry:

> ...add first drop of titrant, contents of reaction flask turn pink immediately, then after few seconds contents turn clear. add second drop of titrant, contents turn pink and immediately turn clear.

Odd. It's as if the reaction rate started off slow after adding the first drop of titrant, then suddenly became faster after the second drop. Weird. You'd normally expect the reaction rate to steadily drop as the titration

went on, given the concentration of ethanol in the vodka is becoming lower and lower throughout the titration. Any theories as to why?

Again, a clue is given in the research the student made:

Mn^{2+} can act as a catalyst for the reduction of MnO_4^- ions.

A catalyst, huh? We know catalysts increase reaction rate. Oh wait - Mn^{2+} looks a little familiar...let's have a look at the reaction equation.

$$4\ MnO_4^-\ (aq) + 5\ C_2H_5OH\ (aq) + 12\ H^+\ (aq) \rightarrow 4\ \mathbf{Mn^{2+}\ (aq)} + 5\ CH_3COOH\ (aq) + 11\ H_2O\ (l)$$

Well. So the product of the reaction is...also the catalyst for the reaction. That makes sense...so when you add the first drop, originally there is no Mn^{2+} in the reaction flask - it is produced after you add the first drop. Since originally there is no Mn^{2+} in the flask, the reaction would be slow. Then, after adding the second drop - yes, the concentration of ethanol is ever so slightly lower, **BUT** you have some Mn^{2+} which acts as a catalyst, which would massively increase the reaction rate!

Another interesting observation - did you notice that the observations in titration 3 look very weird?

...titrant added as a stream initially. contents of reaction flask turn pink immediately, after a few seconds contents turn clear. brown solid observed on bottom of the flask. otherwise, observations as per titration #1

What in the world is this brown solid? Well - if you look at the question stem, it states that the student found through research that MnO_2 is a brown solid.

Also, it just so happens that in titration #3, the titre volume is significantly higher than in the other titrations? Coincidence much?

Where did this MnO_2 come from? It could only have come from the permanganate (MnO_4^-). I invite you to notice that MnO_2 is an incompletely reduced product. How do you know this? Well, the oxidation number of Mn in MnO_4^- is +7, whereas that in the intended product, Mn^{2+} is +2. The oxidation number of Mn in MnO_2 is in between - +4. So therefore, we can speculate that some of the MnO_4^- was incompletely reduced to MnO_2 in this titration. What does this mean? Well, since some of the MnO_4^- was incompletely reduced, this particular amount of permanganate would have "sucked" fewer electrons from the ethanol in the conical flask, and therefore would have oxidised less ethanol into ethanoic acid than it would have had it been completely reduced to Mn^{2+}. This may account for the elevated titre volume in titration 3 - because you would need more permanganate solution to oxidise all the ethanol.

Therefore, we can write something like this:

However, the accuracy of the data should be called into question; it is quite possible that the experimental technique overestimated the true ethanol content of the vodka. Firstly, it was noted that bubbles were visible during the titrations, which may be attributed to the formation of CO_2 in the conical flask from the oxidation of the product ethanoic acid by permanganate ions. Secondly, the vodka may have contained other reducing agents that could have reacted with the permanganate ions. These phenomena would mean that some of the permanganate added to the flask would not have reacted with ethanol, meaning a greater volume of titrant would have been needed to react with all the ethanol and reach the endpoint. Additionally, given that the endpoint itself occurs when there is excess permanganate in the conical flask, it would occur just after the equivalence point. This issue would be compounded by the lack of use of half-drops in the methodology, which increases the risk of overshooting the endpoint. This would mean that the titre volume would be higher, again increasing the calculated ethanol content in the vodka.
With respect to improvements to the methodology, if CO_2 formation indeed had occurred, this issue could be rectified by the use of a weaker oxidising agent than permanganate ions, one that would not react with ethanoic acid (but would with ethanol). The systematic error caused by the presence of other reducing agents in the vodka could be mitigated by the use of a weaker oxidising agent, but is intrinsic to the volumetric technique and is an accepted limitation of redox titrations in general. The use of half-drops (or even quarter-drops) would decrease the risk of overshooting the endpoint.
Interesting observations noted in the experiment are as follows:

- The reaction rate increased after addition of the first drop of titrant. This is likely due to the formation of Mn^{2+} as a product upon adding the initial drop of titrant, which itself is a catalyst for the reaction.
- A brown solid was formed in titration #3. This is likely MnO_2, which may have been produced from the incomplete reduction of MnO_4^- ions. Since these MnO_4^- ions have been incompletely reduced, a smaller amount of ethanol would have been oxidised by these MnO_4^- ions. This means a greater amount of MnO_4^- ions would have been required to react with all the ethanol, which would explain the very high titre volume obtained.

Conclusion

Now, we need to write a succinct conclusion that answers the aim. In the Introduction section, the aim was stated:

> This experiment aims to determine the alcohol content (present as ethanol) in a sample of Grey Goose vodka via volumetric titration against a standard solution of potassium permanganate.

Let us answer the aim now. What was the alcohol/ethanol content in the vodka? We calculated it to be 42.5% (v/v). Now, is this the true value? Can we be certain of it? Well, it may be the true value. It may not be the true value - we cannot be entirely sure, especially given the potential systematic errors in the experiment. So, in your conclusion, answer your aim - state what you got, how it compares to the manufacturer's value, and how certain you are that your experimentally determined value is accurate:

> The ethanol content in the vodka was determined to be 42.5% (v/v), higher than the manufacturer's quoted value of 40.0% (v/v). However, the accuracy of this finding is questionable, due to the possibility of the methodology employed causing an overestimate of the ethanol content.